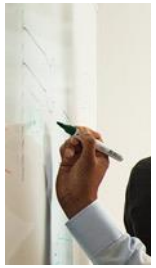
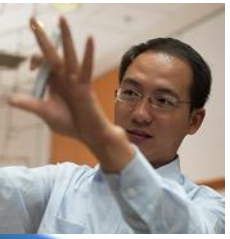
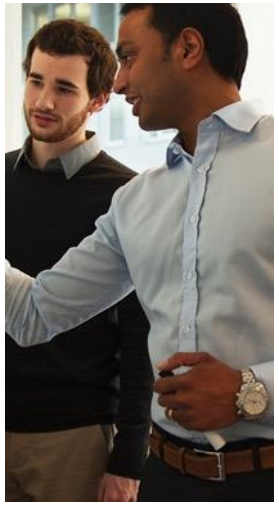






AI Machine Vision

Frequent Asked Questions (FAQ)

TE CONNECTIVITY

Version 2.0

EVERY CONNECTION COUNTS

Contents

1. AIMV Function
    1) What's AI Defect Inspection?
    2) What's Vision Guidance?
    3) What's Dimensional Inspection?
    4) What's the Difference Between AI Defect Inspection, Vision Guidance and Dimensional Inspection?
    5) What's the Advantage of AI Defect Inspection Compared with Traditional Inspection?
    6) What's the Difference Between AIMV, Cognex Vidi and Halcon?
2. Vision Inspection Hardware Selection
    1) What's the Supported 2D Camera Type?
    2) 2D Camera Selection
    3) 3D Camera Selection
    4) Lens Selection
    5) Lighting Selection
3. AIMV Software Installation
    1) What's the Recommended PC Configuration for AIMV Software?
    2) What to Do If the AI Environment Installation Fails?
    3) What to Do If Occur the Pylon Installation Problem?
    4) How to Make Sure the Software Installation for Vision Guidance or Dimensional Measurement is Done?
    5) How to Make Sure the Software Installation for AI Defect Inspection is Done?

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Contents

4. AI Defect Inspection
    1) What's the Difference Between Three AI Defect Inspection Tools?
5. AI Defect Inspection – Step 1 for Image Collection
    1) How to Set Correct Value for "Display Exposure", "High Exposure", "Low Exposure"?
    2) Do I Need to Perform the White Balance?
    3) How to Acquire Images with Good Quality for Training?
    4) How to Stop Real-time Display During Inline Image Collection?
6. AI Defect Inspection – Step 2 for Image Segmentation
    1) Does My Application Need Image Segmentation?
    2) Why can't Segment All the Products Within One Image?
7. AI Defect Inspection – Step 3 for Image Labelling
    1) What's the Rule for Image Labelling?
8. AI Defect Inspection – Step 3 for Image Marking
    1) What's the Rule for Image Marking?

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Contents

8. AI Defect Inspection – Step 4 for Train Model by AWS Server
    1) What's the Relationship Between Product Volume and the Accuracy for Classification?
    2) How to Train One Model Based on One Existing Model?
    3) How to Train One Model Applied for Unaligned Product?
    4) What to Do If the AI Classification Accuracy is Not Good Enough?
    5) How to Decide Epoch Number?
    6) How to Extend the AI Model Training Hours for AWS Training Platform?
9. AI Defect Inspection – Step 5 for Deployment
    1) How to Stop Real-time Display during Inline Deployment?
    2) What to Do If Need a Customized GUI?
10. Calibration
    1) What's Calibration?
    2) How to Define the Conversion Relationship?
    3) Tips for Calibration
    4) What's Auto Calibration?
11. Terminology
12. Contact Us

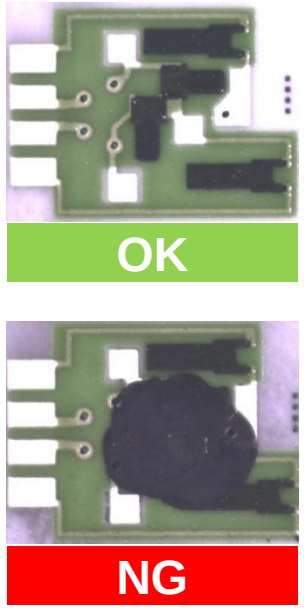
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AIMV Function

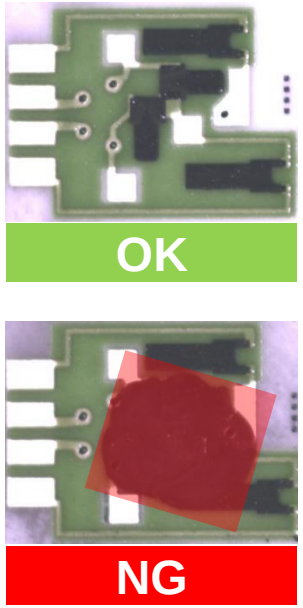
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's AI Defect Inspection?

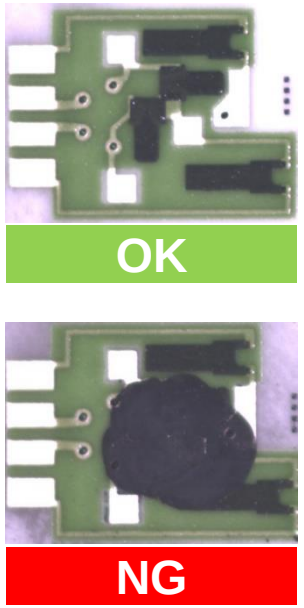
- AI Defect Inspection will use the AI technology to inspect the defects of the product
  - AI Technology: Firstly train one model based on known OK product images and NG product images, then the trained model can work independently to inspect new product
  - Defects: the visible defects such as contamination, discolor, flashing, short-shot, scratches, dents, welding, missing features, etc.
- The functions of AI Defect Inspection include



Classification
(Train AI Model based on OK and NG samples)



Defect Location
(Train AI Model based on marked NG samples)



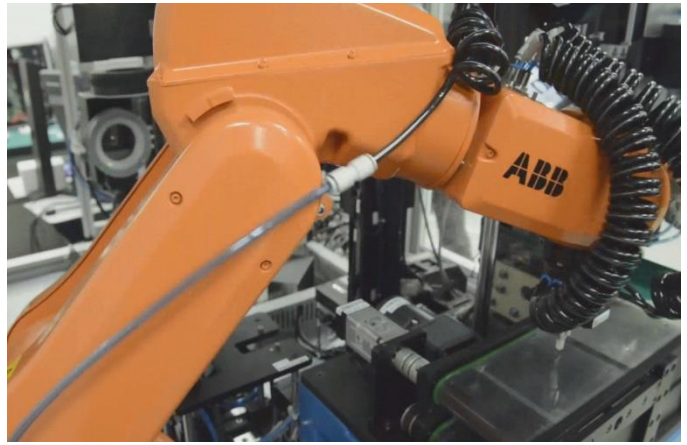
Anomaly Detection
(Train AI Model based on OK samples)

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

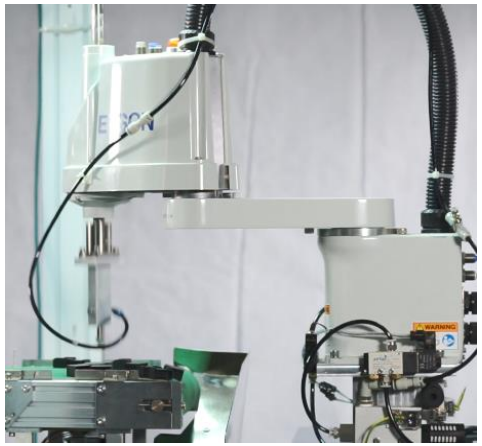
What's Vision Guidance?

- Vision guidance means to guide the executable units to the desired position to perform the required actions
- The executable units include robot, cylinder, motor, etc.
- The required actions include the pick & place, the soldering, the assembly, etc.

Vision Guidance with UR Robot



Vision Guidance with UR Robot



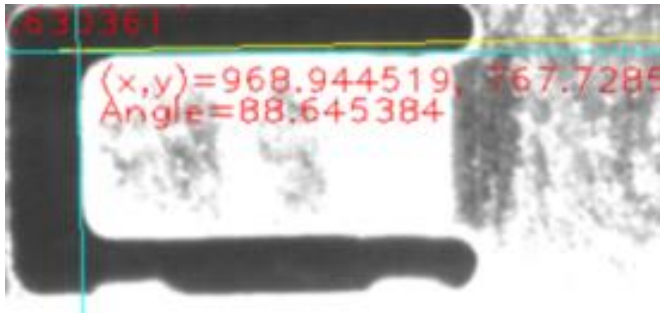
Vision Guidance with EPSON Robot

Vision-guided robots like mounting the eye to the robot, which greatly increases the accuracy of robot movement and enlarges application fields of robot

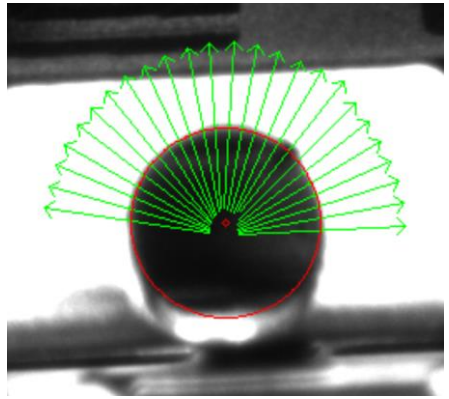
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's Dimensional Inspection?

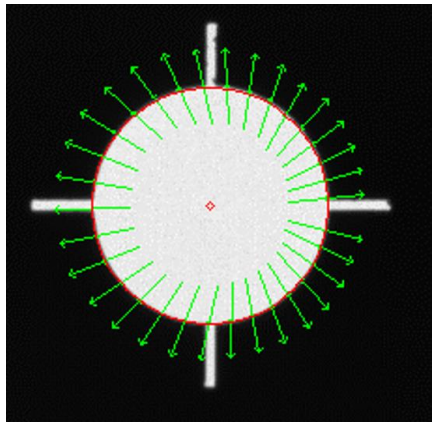
- Dimensional Inspection means to perform quantitative analysis on the feature of the product
- The quantitative features include width, height, area, diameter, angle, pitch, coplanarity, concentricity, position, etc.



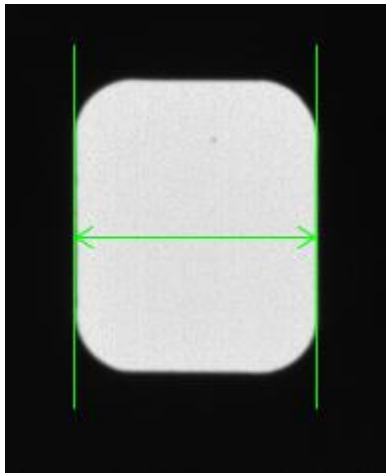
Corner Position, Line angle, Line position



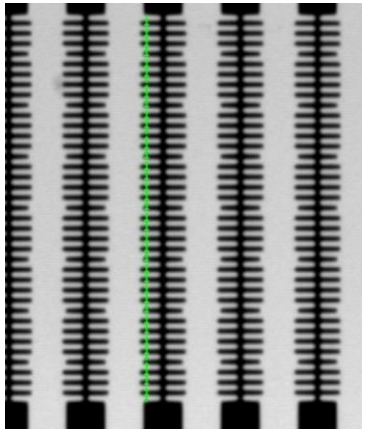
Circle Position, Radius, Area



Diameter and Concentricity



Measure Width



Measure Width & Pitch

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

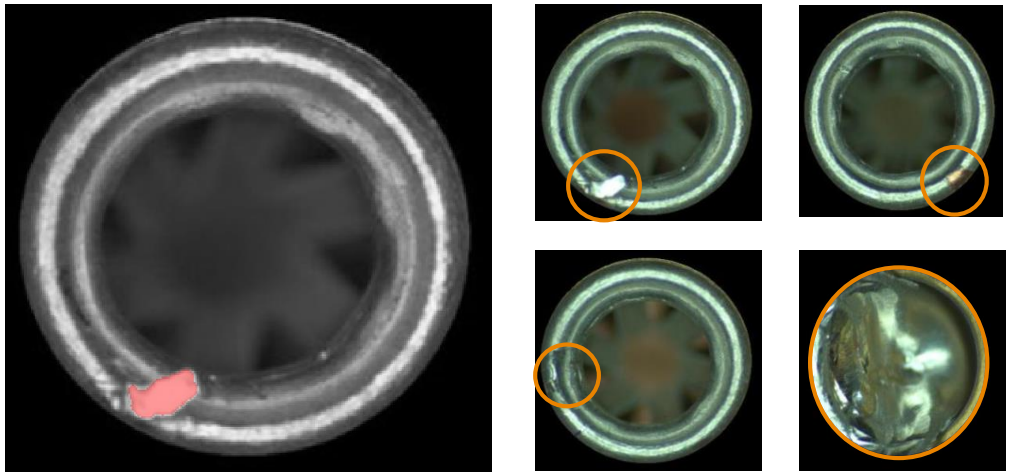
What's the Difference Between AI Defect Inspection, Vision Guidance and Dimensional Inspection?

	Defect Inspection	Vision Guidance	Dimensional Inspection
Applications	Cosmetic Inspection	Robot/Motion-Unit Guidance	Dimensional Measurement
Inline and Offline	Inline and Offline	Inline	Inline and Offline
Need AI Model Training	Yes	No	No

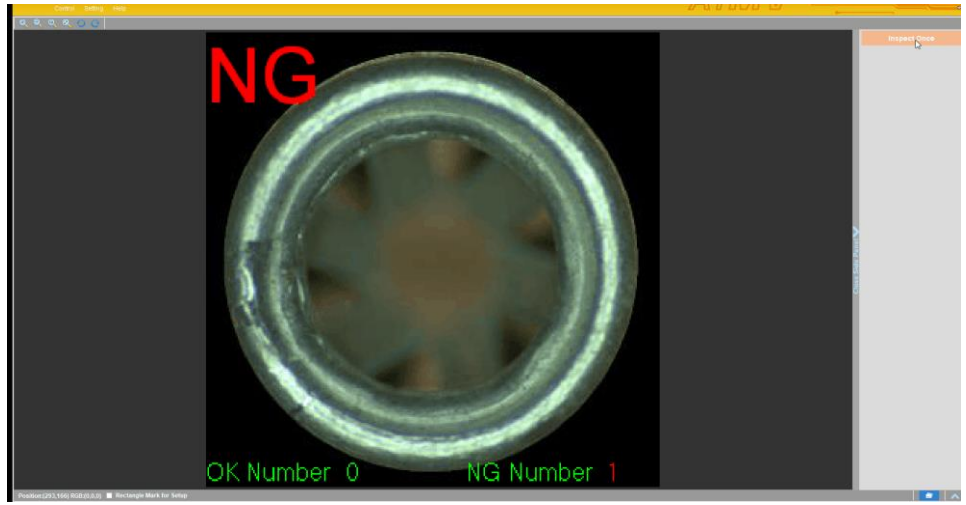
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Advantage of AI Defect Inspection Compared with Traditional Inspection?

- AI Defect Inspection is flexible for the cosmetic Inspection which
  - Has kinds of defects
  - Hard to define defect feature for traditional inspection



Traditional Inspection
(Each Kind of Defects needs One Pre-defined feature)



AI Defect Inspection
(No Predefined Feature Needed)

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Difference Between AIMV, Cognex and Halcon?

Solution	TE AIMV	Cognex ViDi	MVTec Halcon
Software	AIMV (RTD) + AI model cloud training	VisionPro + ViDi	Halcon
Application / Functionality	• Deep Learning Tools: Multiple-Category Classification, Defect Location and Anomaly Detection • Simplest Deployment Process: (1) image collection, segmentation and labeling; (2) Ai modeling training; (3) AI model deployment • Vision Guidance • Dimensional Inspection (Gauging)	• AI Inspection: classify >= two categories by training and deployment phases • Vision Guidance • Dimensional Inspection (Gauging) • Identification (OCR)	• AI Inspection: object detection, classification, segmentation • Vision Guidance • Dimensional Inspection (Gauging) • Identification (OCR)
Preprocessing	Segmentation and labeling	Segmentation and labeling	Segmentation and labeling
Training	Cloud training. Very few parameters	ViDi local training. Many parameters and parameter search	Halcon local training. Need advanced programming
Deployment	Done by 3 Steps in templates. No programming required.	Require Cognex VisionPro + Cognex Designer (or .Net programming)	No standalone deployment tool, need integration to other program for deployment
Amount of training images	>100 images depending on applications	>100 images depending on applications	>100 images depending on applications

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Vision Inspection Hardware Selection

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Supported 2D Camera Type?

- By default, AIMV software supports all Basler acA camera models, please refer to the Basler website https://www.baslerweb.com/en/
- Here is the several commonly-used model for your reference

Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
acA640-100gm	0.3 Mega	659*494	Mono	100	GIGE
acA640-90gc	0.3 Mega	658 * 492	Color	90	GIGE
acA1600-20gm	2 Mega	1626 * 1236	Mono	20	GIGE
acA1920-25gm	2 Mega	1920 * 1080	Mono	25	GIGE
acA1920-150uc	2.3 Mega	1920 * 1200	Color	150	USB 3
acA2500-20gc	5 Mega	2592 * 2048	Color	20	USB 3
acA3800-10gm	10 Mega	3840 * 2748	Mono	10	USB 3
acA4096-11gm	8.8 Mega	4096 * 2160	Mono	11	USB 3

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Supported 2D Camera Type?

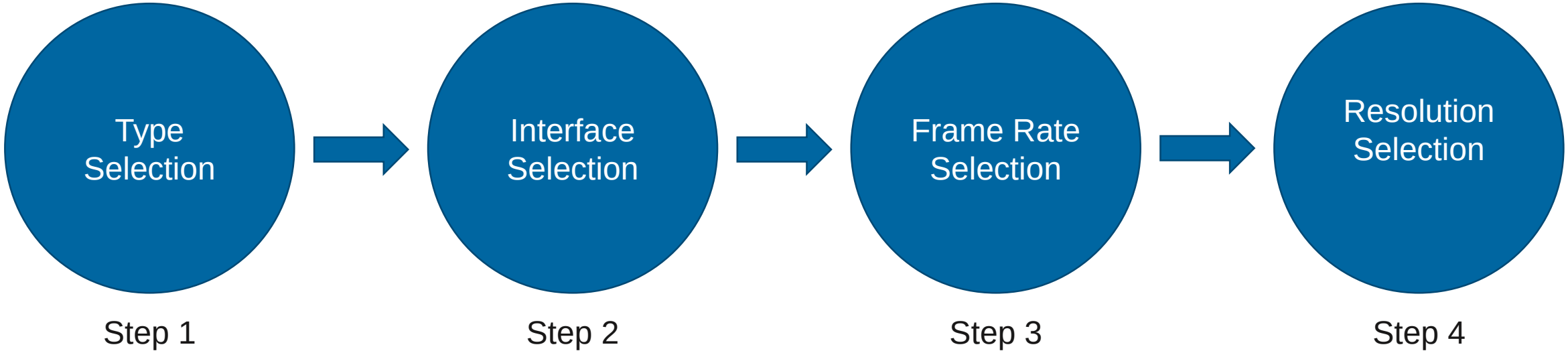
DO3THINK

http://www.do3think.com/?web_lang=en

Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
U3S230M	2.3 Mega	1920 * 1200	Mono	40	USB 3
U3S230	2.3 Mega	1920 * 1200	Color	40	USB 3
U3ST230M	2.3 Mega	1920 * 1200	Mono	40	USB 3
M3S507M	5 Mega	2448 * 2048	Mono	52	USB 3
U3ST507M	5 Mega	2448 * 2048	Mono	52	USB 3
U3P630	6.3 Mega	3072 * 2048	Color	71	USB 3
U3S2000	20 Mega	5472 * 3648	Color	9	USB 3
U3P2000	20 Mega	5480 * 3648	Color	22	USB 3
M3ST2000M	20 Mega	5480 * 3648	Mono	22	USB 3
M3S2000M	20 Mega	5480 * 3648	Mono	22	USB 3
M3S2000	20 Mega	5480 * 3648	Color	22	USB 3

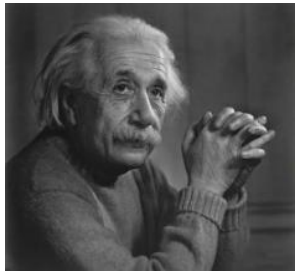
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection



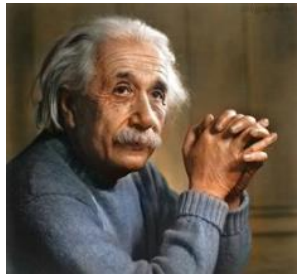
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 1 - Type Selection



Mono

For applications with high accuracy requirement, such as dimensional measurement, vision guidance



Color

For applications to replace visual inspection and need color information

Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
acA640-100gm	0.3 Mega	659*494	Mono	100	GIGE
acA640-90gc	0.3 Mega	658 * 492	Color	90	GIGE
acA1600-20gm	2 Mega	1626 * 1236	Mono	20	GIGE
acA1920-25gm	2 Mega	1920 * 1080	Mono	25	GIGE
acA1920-150uc	2.3 Mega	1920 * 1200	Color	150	USB 3
acA2500-20gc	5 Mega	2592 * 2048	Color	20	USB 3
acA3800-10gm	10 Mega	3840 * 2748	Mono	10	USB 3
acA4096-11gm	8.8 Mega	4096 * 2160	Mono	11	USB 3

*The table is for illustration. Refer to FAQ “Supported Camera Type” for supported camera type

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 2 - Interface Selection

USB

- The maximum transitions distance: 5 meters
- The maximum transitions bandwidth: 400 MB/S
- Do not need extra hardware for PC

GIGE

- The maximum transitions distance: 100 meters
- The maximum transitions bandwidth: 100 MB/S
- Need Extra Ethernet card for PC

Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
acA640-100gm	0.3 Mega	659*494	Mono	100	GIGE
acA640-90gc	0.3 Mega	658 * 492	Color	90	GIGE
acA1600-20gm	2 Mega	1626 * 1236	Mono	20	GIGE
acA1920-25gm	2 Mega	1920 * 1080	Mono	25	GIGE
acA1920-150uc	2.3 Mega	1920 * 1200	Color	150	USB 3
acA2500-20gc	5 Mega	2592 * 2048	Color	20	USB 3
acA3800-10gm	10 Mega	3840 * 2748	Mono	10	USB 3
acA4096-11gm	8.8 Mega	4096 * 2160	Mono	11	USB 3

*The table is for illustration. Refer to FAQ “Supported Camera Type” for supported camera type

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 3 – Max Frame Rate Selection

Basic Knowledge

- **Cycle Time of Vision Application:** Image Capture Time + Image Processing Time + Communication Time
- **Image Capture Time**: 1000 / Actual Frame Rate
- **Max Frame Rate:** at most how many images the camera can capture per second. The actual frame rate is influenced by the exposure value. To achieve Max Frame Rate, the exposure value should be smaller than 1000 / Max Frame Rate

- Please select one camera whose Max Frame Rate satisfies the Cycle Time Requirement of your Vision Application

Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
acA640-100gm	0.3 Mega	659*494	Mono	100	GIGE
acA640-90gc	0.3 Mega	658 * 492	Color	90	GIGE
acA1600-20gm	2 Mega	1626 * 1236	Mono	20	GIGE
acA1920-25gm	2 Mega	1920 * 1080	Mono	25	GIGE
acA1920-150uc	2.3 Mega	1920 * 1200	Color	150	USB 3
acA2500-20gc	5 Mega	2592 * 2048	Color	20	USB 3
acA3800-10gm	10 Mega	3840 * 2748	Mono	10	USB 3
acA4096-11gm	8.8 Mega	4096 * 2160	Mono	11	USB 3

*The table is for illustration. Refer to FAQ “Supported Camera Type” for supported camera type

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 4 - Resolution Selection

1. The resolution of camera is calculated by the application
2. Please calculate the required resolution based on the steps in the following slides
   1) AI Defect Inspection
   2) Dimensional Inspection
   3) Vision Guidance
3. Select the camera with resolution which is bigger than required resolution

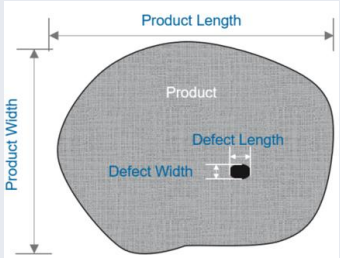
Model	Resolution	Width * Height	Type	Max Frame Rate (FPS)	Interface
acA640-100gm	0.3 Mega	659*494	Mono	100	GIGE
acA640-90gc	0.3 Mega	658 * 492	Color	90	GIGE
acA1600-20gm	2 Mega	1626 * 1236	Mono	20	GIGE
acA1920-25gm	2 Mega	1920 * 1080	Mono	25	GIGE
acA1920-150uc	2.3 Mega	1920 * 1200	Color	150	USB 3
acA2500-20gc	5 Mega	2592 * 2048	Color	20	USB 3
acA3800-10gm	10 Mega	3840 * 2748	Mono	10	USB 3
acA4096-11gm	8.8 Mega	4096 * 2160	Mono	11	USB 3

*The table is for illustration. Refer to FAQ “Supported Camera Type” for supported camera type

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

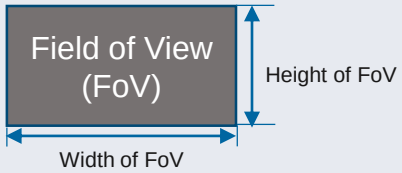
2D Camera Selection: Step 4 - Resolution Calculation for AI Inspection

1 Have Product Size and Minimum Defect Size



AI Inspection Example:
- Product Size = 40 * 30 mm
- Minimum Defect Size = 0.3 * 0.3 mm

2 Calculate the Field of View (FoV) of Your Application



- FoV is how large area of one image in the real world

AI Inspection Example:
- FoV = 150 mm * 100 mm

3 Required Resolution

$$Width = Max\left(\frac{Width\ of\ FoV}{Width\ of\ Product} * 299, \frac{Width\ of\ FoV}{Width\ of\ Defect} * 6\right)$$

$$Height = Max\left(\frac{Height\ of\ FoV}{Heiht\ of\ Product} * 299, \frac{Heigh\ of\ FoV}{Heigh\ of\ Defect} * 6\right)$$

AI Inspection Example:
- $Width = Max\left(\frac{150}{40} * 299, \frac{150}{0.3} * 6\right)$ = 3000
- $Heigh = Max\left(\frac{100}{30} * 299, \frac{100}{0.3} * 6\right)$ = 2000

4 Find Available Camera

Check chapter “Supported Camera Type” to have available camera resolution

Dimensional Measurement Example:
- Result: need at least 10-mega-pixel camera

- By default, for AI Defect Inspection Standard Solution, the camera is Do3Think U3P2000, which is one 20-Mega-pixel colorful camera
- Please use “Resolution Calculation.xlsx” on calculation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 4 - Resolution Calculation for AI Inspection

Calculation Excel

Microsoft Excel Worksheet

Input Paras

Calculated Result

Please Input Information Here					
Product Size		Minimum Defect Size		Field of View	
Width (mm)	Height (mm)	Width (mm)	Height (mm)	Width (mm)	Height (mm)
40	30	0.3	0.3	150	100

Required Resolution	
Width (Pixel)	Height (Pixel)
3000	2000

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Return to Contents

2D Camera Selection: Step 4 - Resolution Calculation for Dimensional Measurement

1 Have Tolerance of Your Application

$$Tolerance = UCL - LCL$$

- UCL (Upper Control Limit)
- LCL (Lower Control Limit)

Dimensional Measurement Example:
- UCL = 4.1 mm
- LCL = 3.9 mm
- Tolerance = 4.1 - 3.9 = 0.2 mm

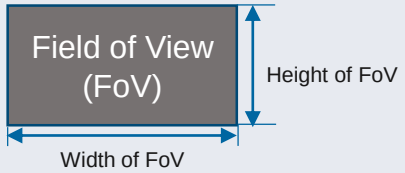
2 Calculate Accuracy Requirement of the Application

$$Accuracy\ Requirement = \begin{cases} Tolerance\ /\ 10 \\ Tolerance\ /\ 5 \end{cases}$$

- If your application need to pass GR&R or similar criteria, please select 10

Dimensional Measurement Example:
- Accuracy Requirement = 0.2 mm / 10 = 0.02 mm

3 Calculate the Field of View (FoV) of Your Application



- FoV is how large area of one image in the real world

Dimensional Measurement Example:
- The product size is 10 * 8 mm
- The field of view is a little bigger to cover one product. For example 12 * 10 mm

4 Estimate the Image Quality Factor of Product

$$Factor = \begin{cases} 3 & Sharp\ Edge \\ 5 & Moderate\ Edge \\ 10 & Blur\ Edge \end{cases}$$

- When lighting, if the image is with sharp edge, select 3, if with blur edge, select 10

Dimensional Measurement Example:
- Image Quality Factor = 3 for lighting image with sharp edge

5 Calculate the Required Resolution

$$Width = \frac{Width\ of\ FoV}{Accuracy\ Requirement} * Factor$$

$$Height = \frac{Height\ of\ FoV}{Accuracy\ Requirement} * Factor$$

Dimensional Measurement Example:
- Width = 12 / 0.02 * 3 = 1800
- Height = 10 / 0.02 * 3 = 1500

6 Find Available Camera

Check chapter “Supported Camera Type” to have available camera resolution

Dimensional Measurement Example:
- Result: need at least 5-mega-pixel camera

- Please use “Resolution Calculation.xlsx” on calculation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 4 - Resolution Calculation for Dimensional Measurement

Calculation Excel

Microsoft Excel Worksheet

Input Paras

Calculated Result

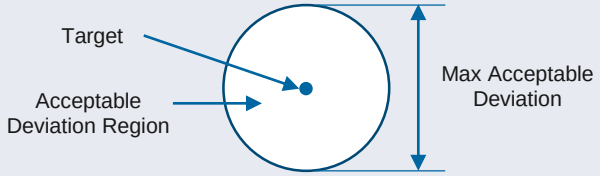
Please Input Information Here				
Tolerance (mm)	Accuracy Requirement (5 or 10)	Field of View		Image Quality Factor (3 or 5 or 10)
		Width (mm)	Height (mm)	
0.2	10	12	10	3

Required Resolution	
Width (Pixel)	Height (Pixel)
1800	1500

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 4 - Resolution Calculation for Vision Guidance

1 Have Max Acceptable Deviation of Your Application



- If exceed the Max Acceptable Deviation, the vision guidance will fail

Vision Guidance Example:
- Max Allowed Guidance Deviation = 0.4 mm

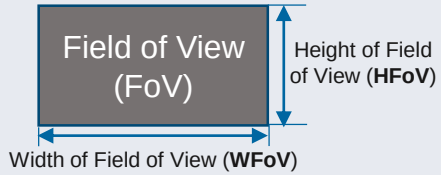
2 Calculate Accuracy Requirement of the Application

$$Accuracy\ Requirement = \begin{cases} Tolerance\ /\ 5 \\ Tolerance\ /\ 3 \end{cases}$$

- If your application has some mechanical fixture to help guidance, such as chamfer, we can select 3

Vision Guidance Example:
- Accuracy Requirement = 0.4 mm / 3 = 0.133 mm

3 Calculate the Field of View (FoV) of Your Application



- FoV is how large area of one image in the real world

Vision Guidance Example:
- The field of view is 50 * 35 mm

4 Estimate the Image Quality Factor of Product

$$Factor = \begin{cases} 3 & Sharp\ Edge \\ 5 & Moderate\ Edge \\ 10 & Blur\ Edge \end{cases}$$

- When lighting, if the image is with sharp edge, select 3, if with blur edge, select 10

Vision Guidance Example:
- Image Quality Factor = 5 for lighting image with moderate edge

5 Calculate the Required Resolution

$$Width = \frac{Width\ of\ FoV}{Accuracy\ Requirement} * Factor$$

$$Height = \frac{Height\ of\ FoV}{Accuracy\ Requirement} * Factor$$

Vision Guidance Example:
- Width = 50 / 0.133 * 5 = 1879
- Height = 35 / 0.133 * 5 = 1315

6 Find Available Camera

Check chapter “Supported Camera Type” to have available the camera resolution

Vision Guidance Example:
- Result: need at least 5-mega-pixel camera

- Please use “Resolution Calculation.xlsx” on calculation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Step 4 - Resolution Calculation for Vision Guidance

Calculation Excel

Microsoft Excel Worksheet

Input Paras

Calculated Result

Please Input Information Here				
Max Acceptable Deviation (mm)	Accuracy Requirement (5 or 3)	Field of View		Image Quality Factor (3 or 5 or 10)
		Width (mm)	Height (mm)	
0.4	3	50	35	5

Required Resolution	
Width (Pixel)	Height (Pixel)
1875	1312.5

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

2D Camera Selection: Selection for Special Requirement

Tip:

- If have special requirement on camera, such as imaging moving product, fast cycle time, etc. please contact us for technical support

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

3D Camera Selection

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Lens Selection

Standard

Select Lens for Standard AI Defect Inspection Solution

Microsoft Excel Worksheet

- Order standard AI Defect Inspection Solution here
- The Lens is adaptive to the Do3Think U3P2000 camera, which is one 20-mega-pixel colorful camera

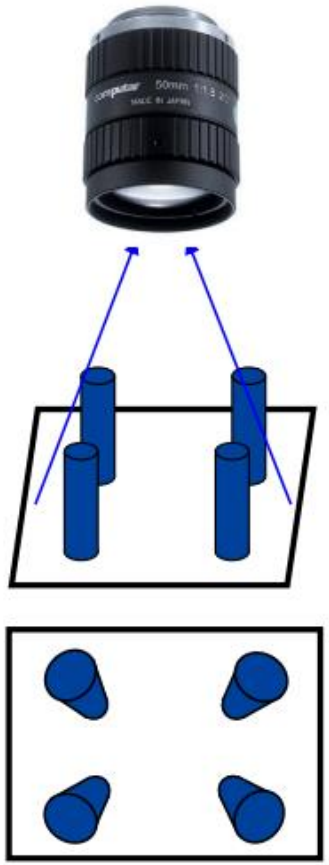
Customized

Select Lens for Application with Customized Configuration

- please contact us for technical support

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lens Selection – FA lens vs Telecentric Lens

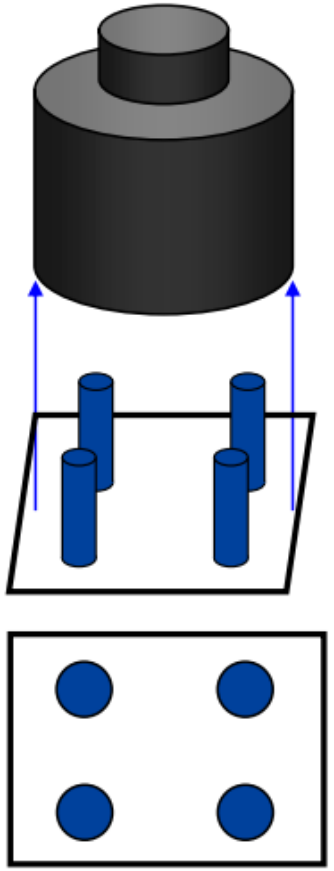


FA Lens

- With perspective
- Low cost
- Small lens size
- Widely used in various applications

Perspective

- Size distortion in different distances
- The farer the distance is, the smaller the product is



Telecentric Lens

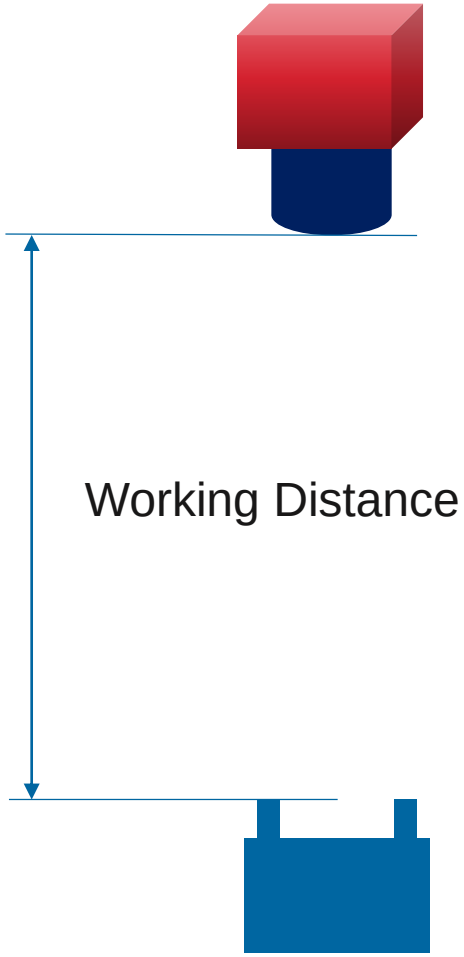
- No perspective
- Higher cost
- Big lens size
- Good image quality
- Suitable for applications with high accuracy requirement, such as dimensional inspection

No Perspective

- The same product size at different distance

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lens Selection – Working Distance



Working Distance (WD)

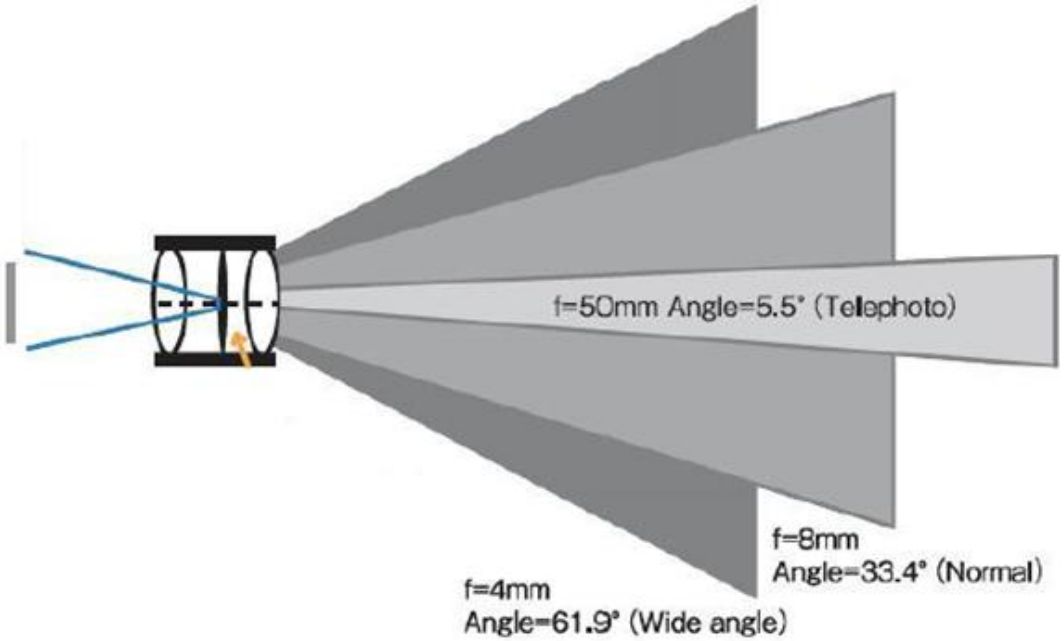
- From the bottom of the lens to the imaging surface of the product

Tips

- WD will influence the mechanical installation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lens Selection – Focal Length



Focal Length

- 4 mm
- 8 mm
- 50 mm

Tips

- Bigger focal length, shorter working distance, smaller FoV

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Lens Selection for Standard AI Defect Inspection Solution

Selection Excel

Microsoft Excel Worksheet

		User Inputs
Each Product Size (mm)	Length	10.3
	Width	7.4
Minimum Defect Size in Each Product (mm)	Length	3
	Width	0.4

							Computing Results
Calculated FoV	Length						112.7
	Width						75.8
Available Lens Types	A12	Working Distance	230~690	Length of FoV	250~750	Width of FoV	167~500
	A16		180~305		145~250		97~167
	A25		230~280		120~145		80~97
	A35		170~320		63~120		42~80
	T52		200		52		35
	T48		200		48		32
	T44		200		44		30
	T37		110		37		25
	T36		200		36		24
	T32		110		32		22
	T30		110		30		20
	T26		220		26		18
	T24		220		24		16
	T22		80		22		15
	T16		65		16		11
	T12		65		13		8.8
	T6		65		6.5		4.9

1

Input
- Product Size
- Minimum Defect Size in Each Product

2

Result
- The suitable lens type will be checked
- Please see whether the working distance is suitable for your application. If not, please contact us for technical support
- Lens Type: A is FA Lens, T is Telecentric Lens

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Lighting Unit Selection

Standard

Select Lighting Unit for Standard AI Defect Inspection Solution

Microsoft Excel Worksheet

- Order standard AI Defect Inspection Solution here
- The Lighting Unit is adaptive to the Do3Think U3P2000 camera, which is one 20-mega-pixel colorful camera

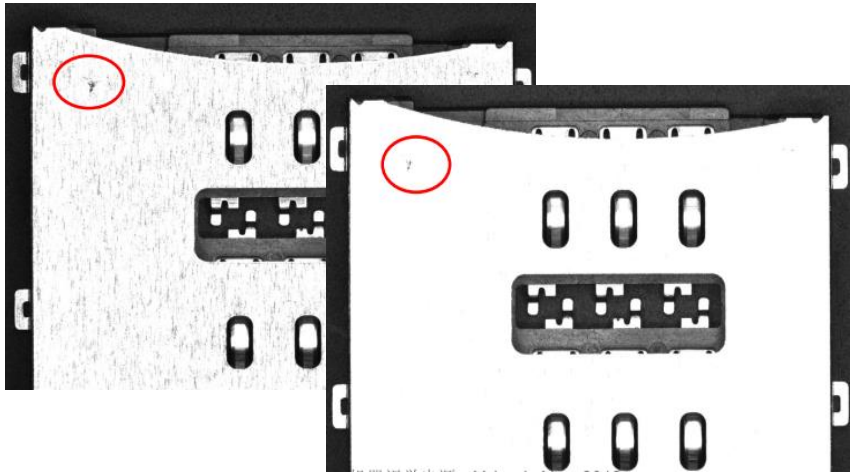
Customized

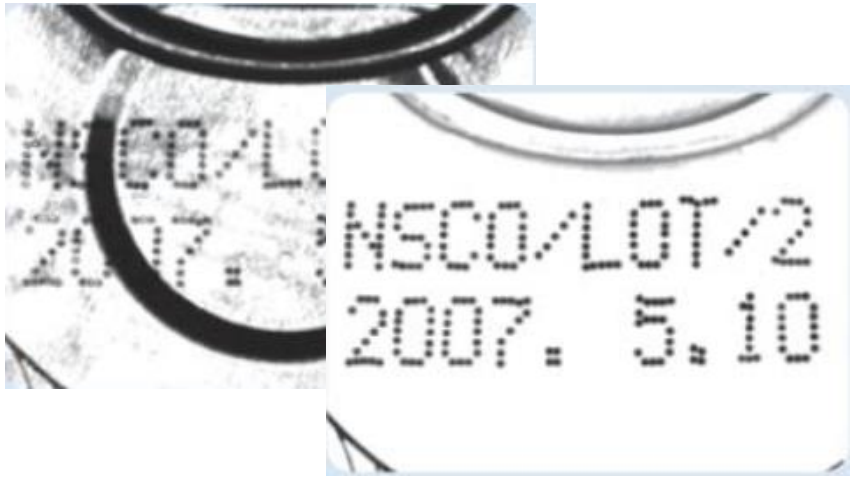
Select Lighting Unit for Application with Customized Configuration

- please contact us for technical support

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Purpose of Lighting Unit



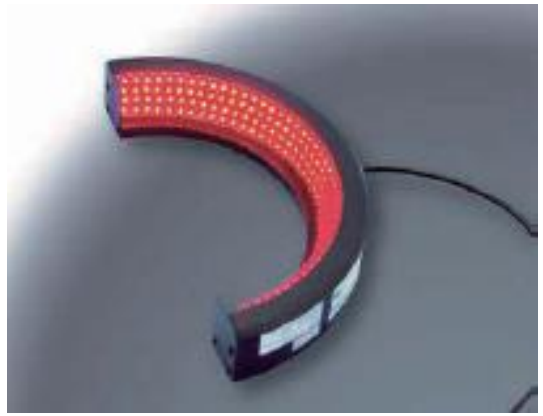


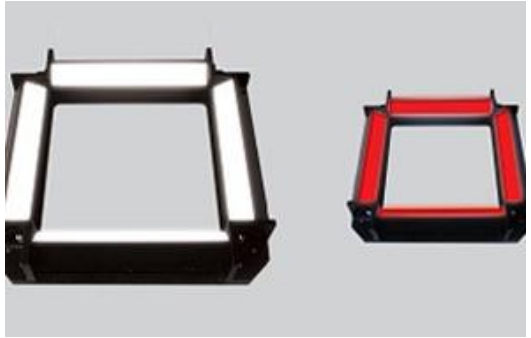
Purpose

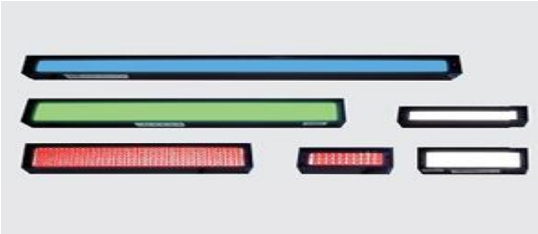
- Increase the contrast to visualize the feature
- Suppress the noise

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

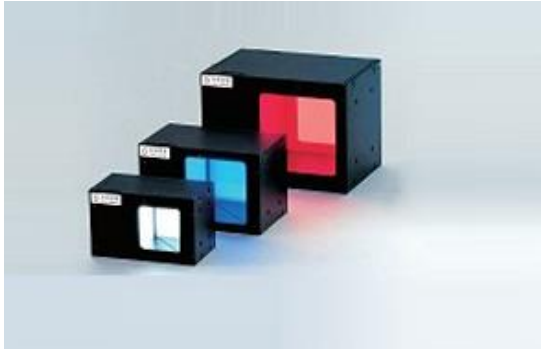
Background Knowledge for Lighting Unit Selection – Different Lighting Units

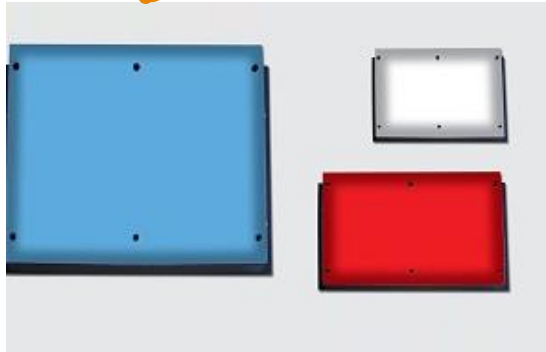






Different lighting sources

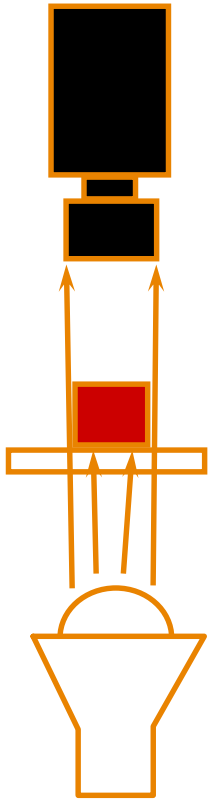


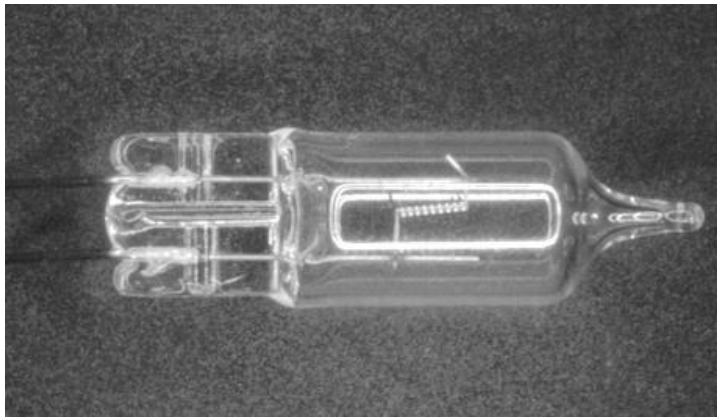


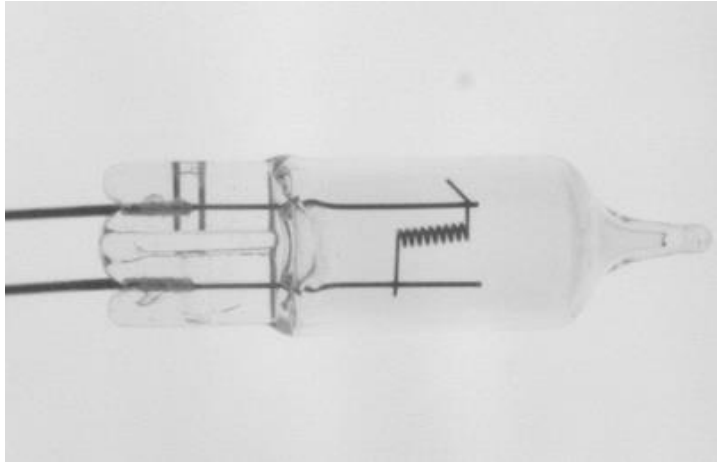
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Back Lighting

- Dimensional
- Penetrate the transparent object



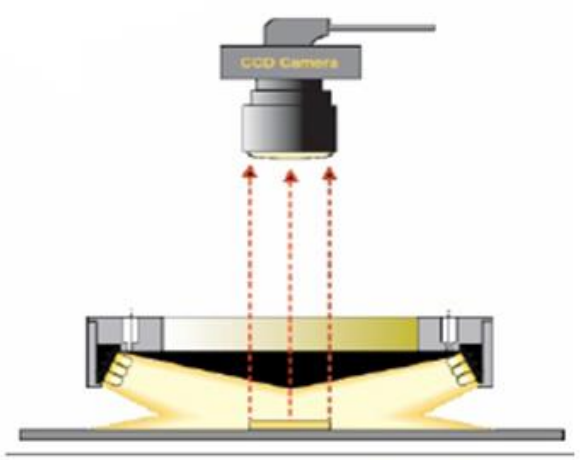




TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

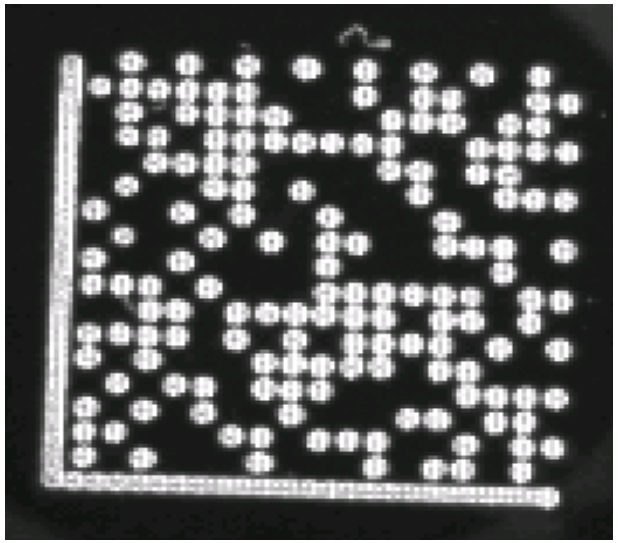
Background Knowledge for Lighting Unit Selection – Front Low Angle Ring Lighting

- Contour
- Shape of surface
- Dark background, bright feature



TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

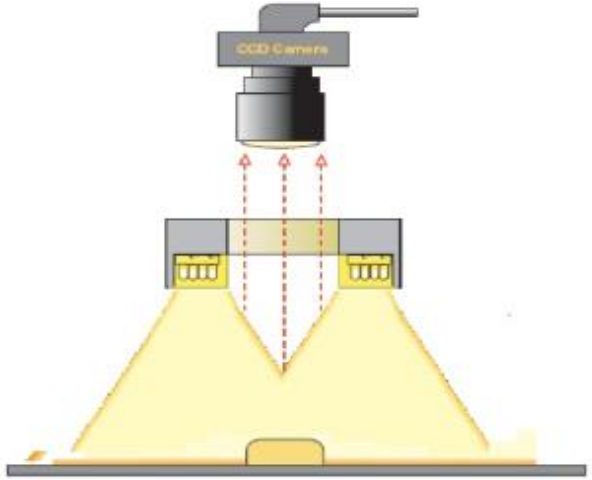
Background Knowledge for Lighting Unit Selection – Front Low Angle Ring Lighting



TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Front High Angle Ring Lighting

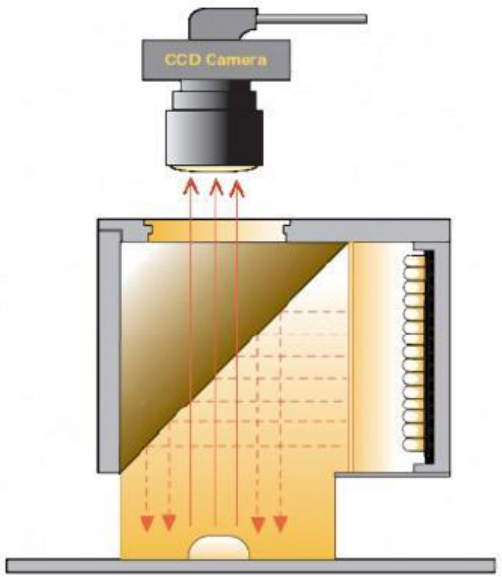
- Even, soft lighting
- Reduce the shadow
- Reduce the contrast
- Adjust height, angle
- Flat object with low reflection

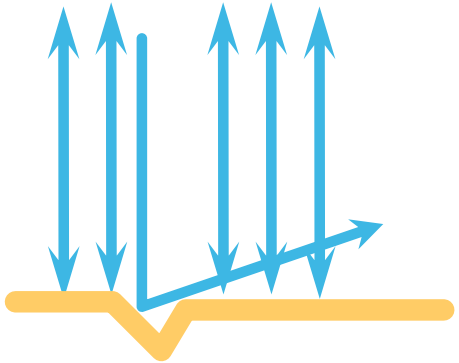


TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Front Coaxial Lighting

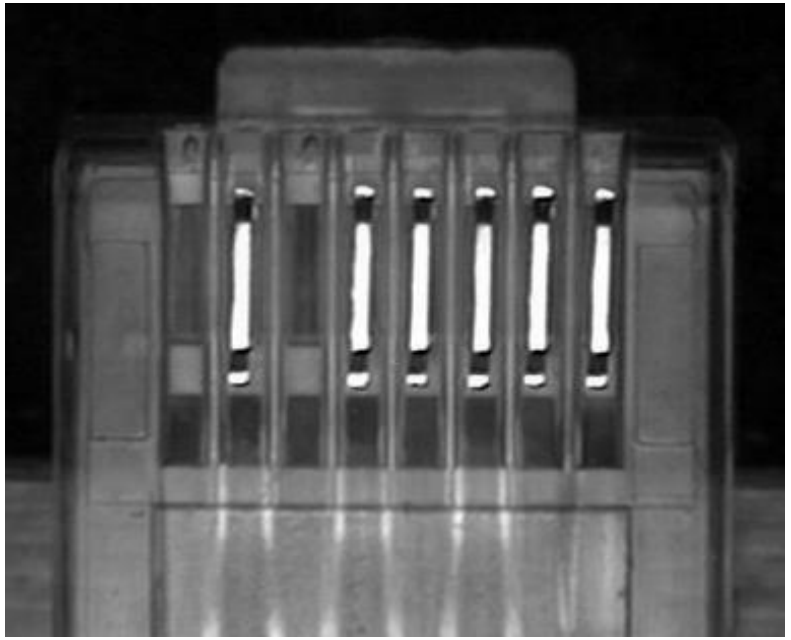
- Parallel lighting
- Mirror surface
- Only for flat object





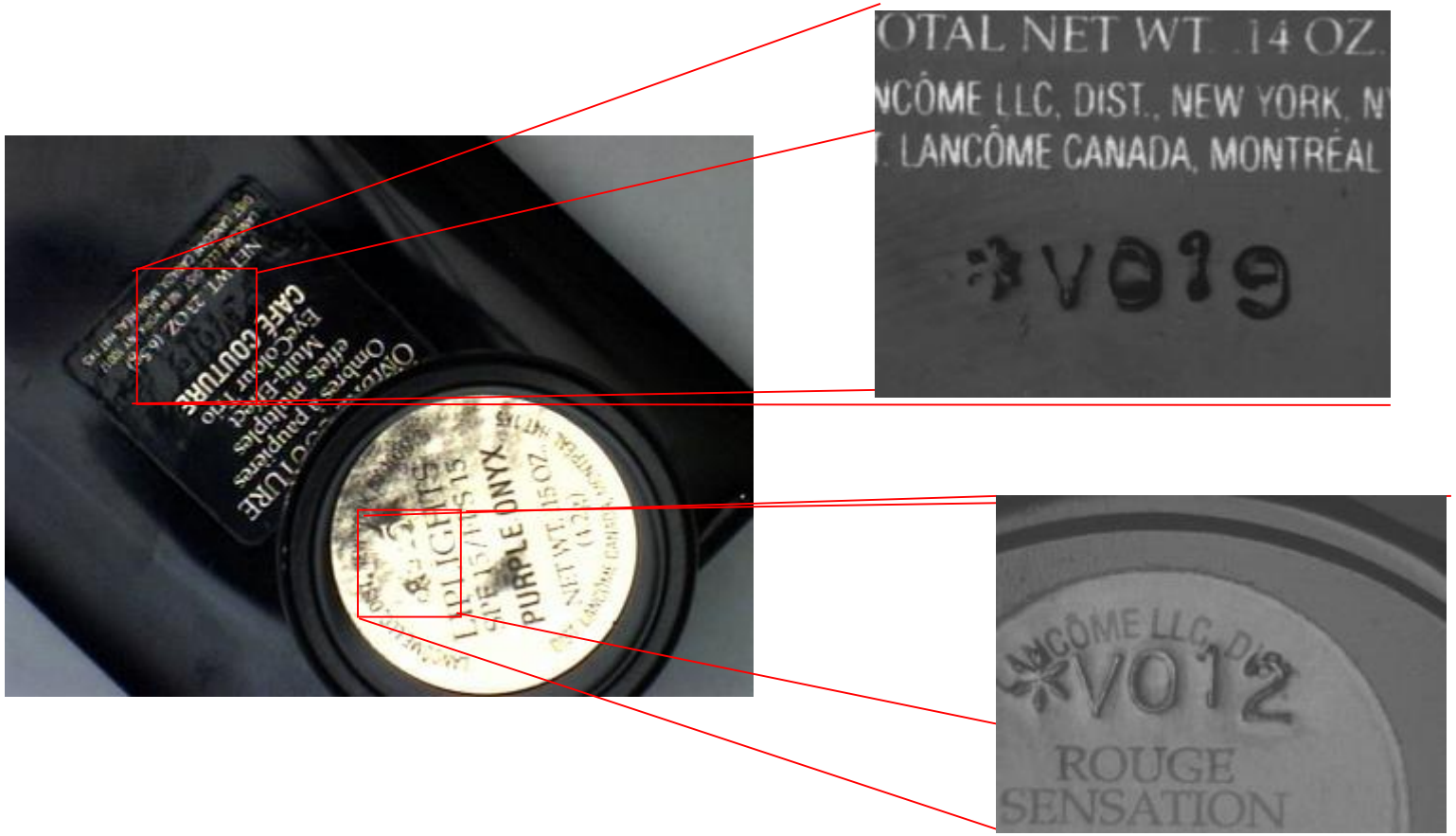
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Front Coaxial Lighting



TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Front Coaxial Lighting



TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Background Knowledge for Lighting Unit Selection – Front Dome Lighting

- Even lighting
- No shadow
- Curved surface

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Lighting Unit Selection for Standard AI Defect Inspection Solution

Selection Excel

Microsoft Excel Worksheet

1

Input
- Product Size
- Minimum Defect Size in Each Product

2

Result
- The suitable lens type will be checked
- Please see whether the lighting unit type is suitable for your application. If not, please contact us for technical support

	User Inputs
FoV Length of Selected Lens	50
Light Type Number (Input 1, 2 or 3)	1

- 1: Front Ring Light (for Surfacial Discoloration & Contamination)
  2: Back Light (for Edge Defect, eg. Short Shot)
  3: Co-Axial Front Light (for Surfacial Scratches & Dent)

				Computing Results
Front Ring Light Type	F3506	Size (Diameter/mm)	350	
	F2726		272	
	F1806		180	√
	F1506		150	
	F906		90	

				Computing Results
Back Light Type	B300300	Size (mm*mm)	300*300	
	B232212		232*212	
	B145131		145*131	
	B125103		125*103	
	B102102		102*102	
	B8060		80*60	

				Computing Results
Co-Axial Front Light Type	C250	Size (Diameter/mm)	250*250	
	C200		200*200	
	C150		150*150	
	C100		100*100	
	C70		70*70	

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AIMV Software Installation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Recommended PC Configuration for AIMV Software?

	AI Defect Inspection	Vision Guidance	Dimensional Inspection
CPU	Intel i5 8th series or better		
GPU	NVIDIA GTX 2060 6G or better	No requirement	No requirement
ROM	>= 16 G		
Volume	>= 512 G, SSD is preferred		
USB	• USB 3.0 • USB ports > 3		
Operation System	Windows 10 and above, 64 bit		
Display	• Screen >= 21 inches • Resolution >= 1080P • Color display		

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What to Do If the AI Environment Installation Fails?

Error

- The AI_Environment Installation fails

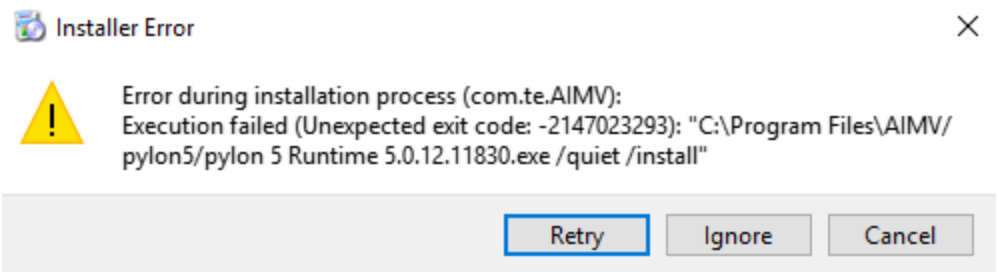
Solution

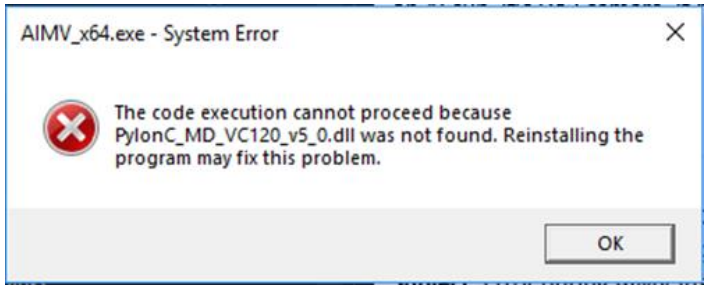
1. Copy “AIMV Package” folder to C:/ in your computer
2. Uninstall any Python related program in your computer, such as
   - Python
   - Anaconda
3. Search Python in your disk to see any remain of the uninstalled software. Delete the remains if you find it
4. Make sure upgrade your NVIDIA graphic card driver to the latest version
5. Restart your PC and reinstall the AI_Environment software
6. If it still doesn’t work, please change your computer or re-install your Win10 system

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What to Do If the Pylon Installation Problem Happens?

Error





Reason

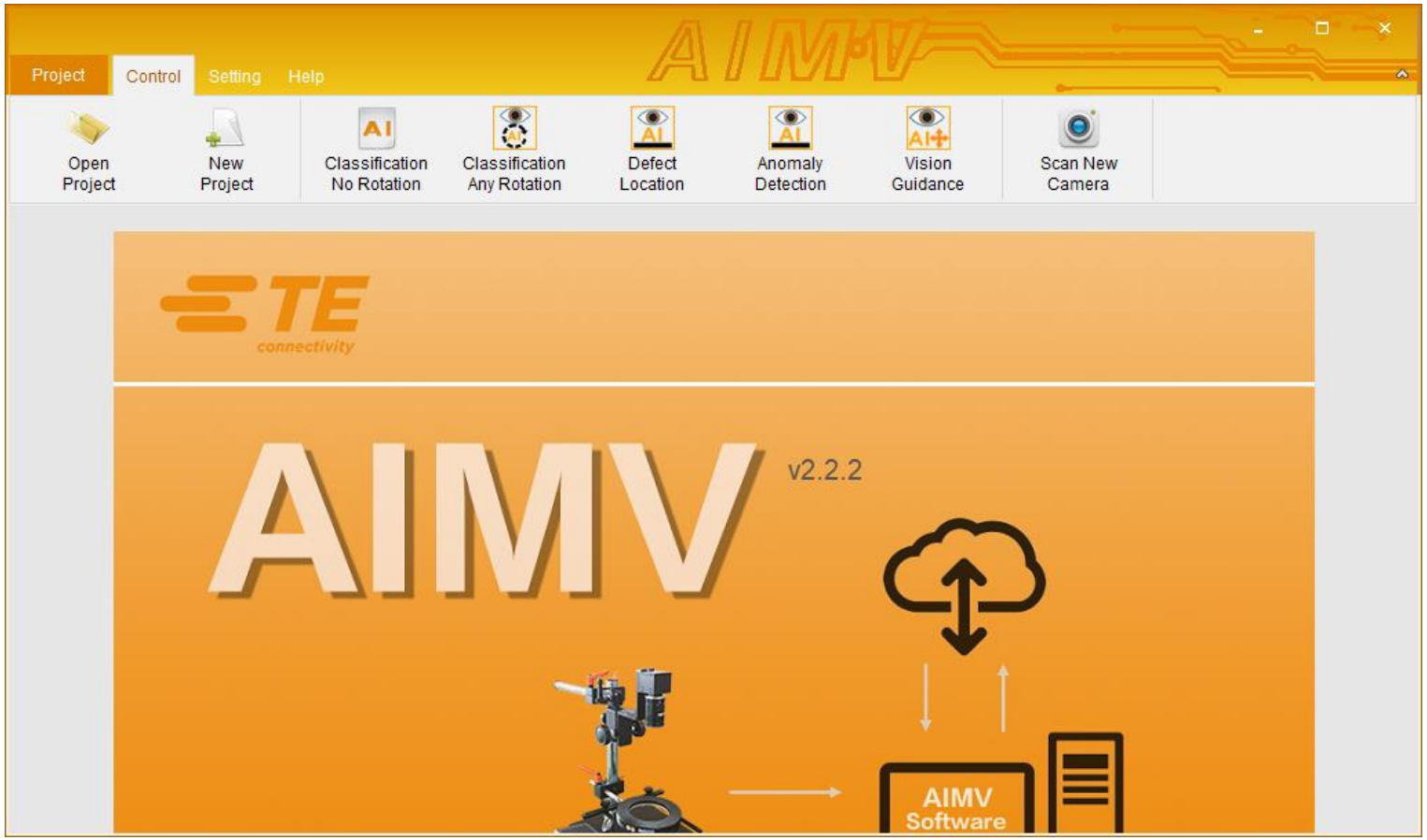
- Pylon is one software package provided by Basler Company. If we can’t install it, it means some software in your system is incompatible to the Pylon Software

Solution

1. Uninstall any software related to Pylon
2. Update your Win10 system to latest
3. Re-install the AIMV software
4. If it still doesn’t work, please change your computer or re-install your Win10 system

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Make Sure the Software Installation for Vision Guidance or Dimensional Measurement is done?

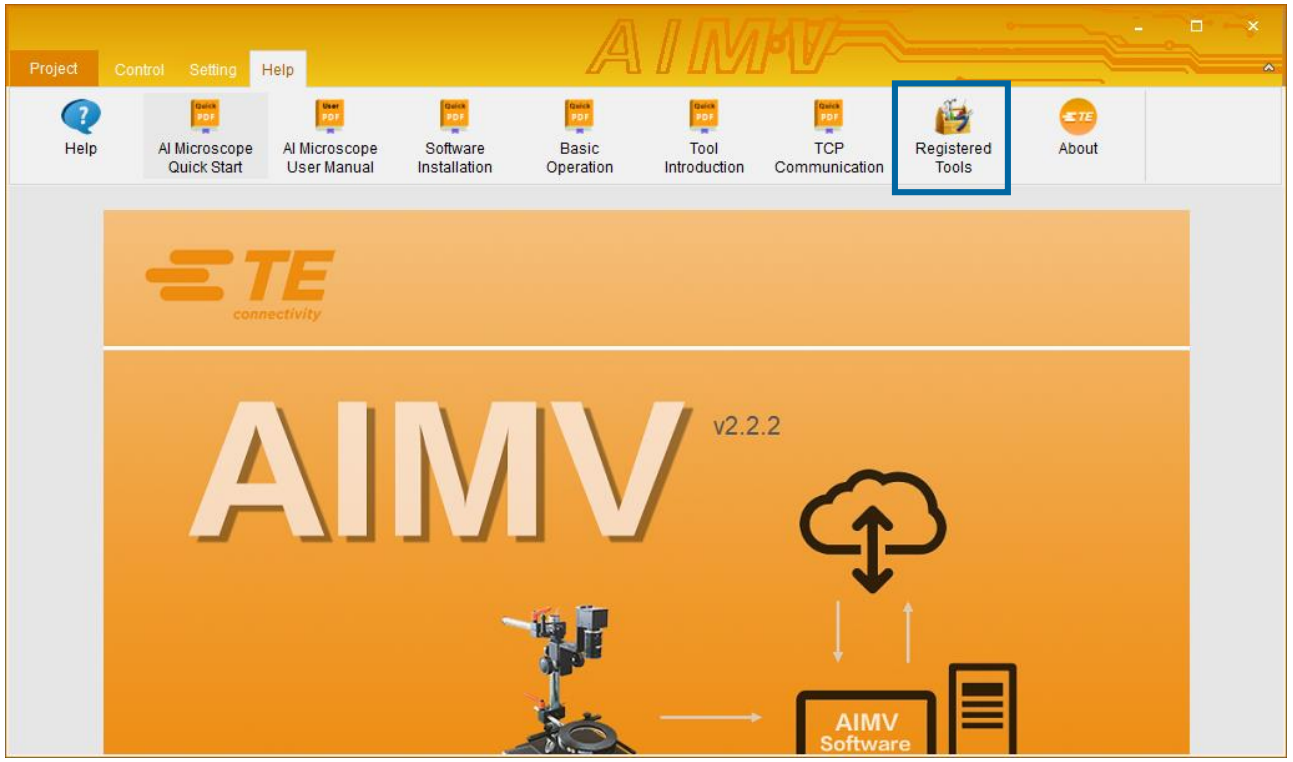


1. Double click AIMV icon to start the software. If popping-up the camera register window, please close the window

2. If the Welcome Page is Popped up, the installation is DONE

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Make Sure the Software Installation for AI Defect Inspection is Done?

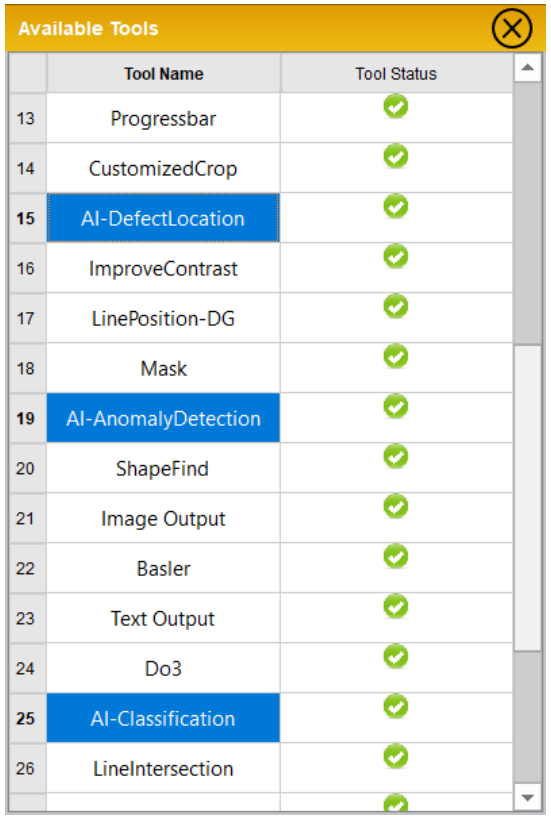


1. Double click AIMV icon to start the software. If popping-up the camera register window, please close the window

2. Switch to “Help Tab” and Click “Registered Tools” button

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Make Sure the Software Installation for AI Defect Inspection is Done?



3. Please make sure “AI-DefectLocation”, “AI-AnomalyDetection” and “AI-Classification” are installed, which means the installation is done.

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AI Defect Inspection

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Difference Between Three AI Defect Inspection Tools?

	AI-Classification	AI-DefectLocation	AI-AnomalyDetection
Classify Product Image to OK/NG	Yes	Yes	Yes
Locate and Display Defect Position	No	Yes	No
Running Speed	☆	☆☆	☆
Training Image	Need OK and NG Product Image	Need NG Image with Marked Defect	Need OK Product
Training Image Volume	Hundreds+	Dozens+	Hundreds+
Manually Mark the Defect	No	Yes	Yes
Input Image Size (Width * Height)	299 * 299	No Limitation	256 * 256

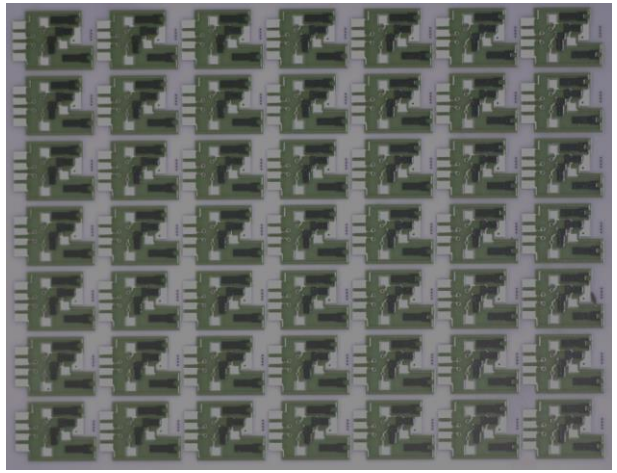
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AI Defect Inspection – Step 1 for Image Collection

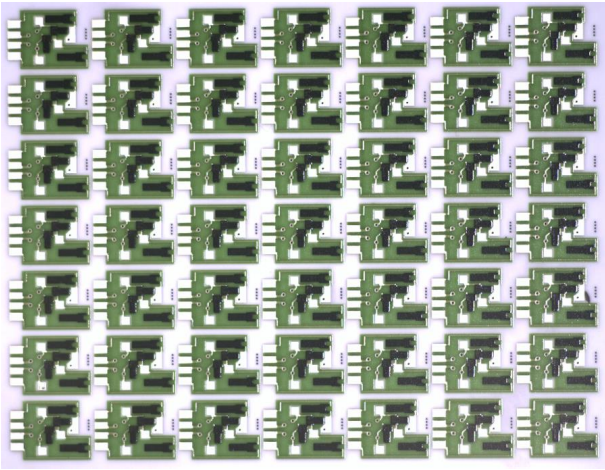
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's Exposure?

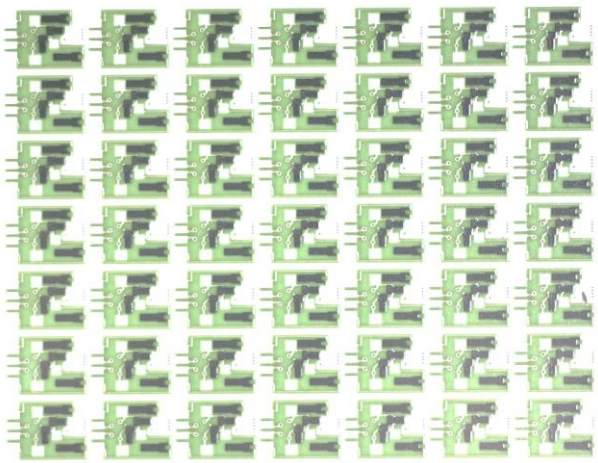
Camera exposure is the overall brightness or darkness of a photograph.



Under Exposed



Normal



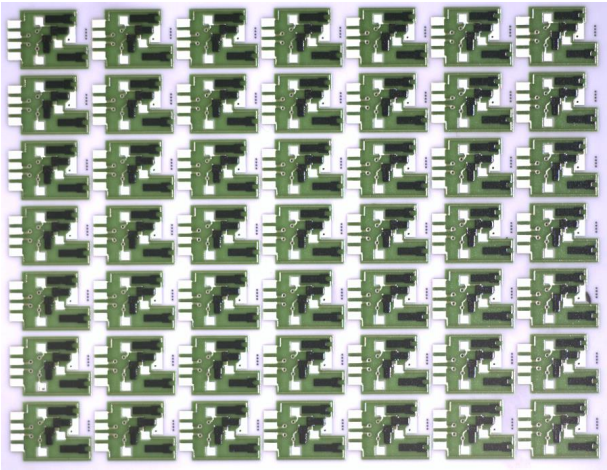
Over Exposed

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Set Correct Value for “Display Exposure”, “High Exposure”, “Low Exposure”?

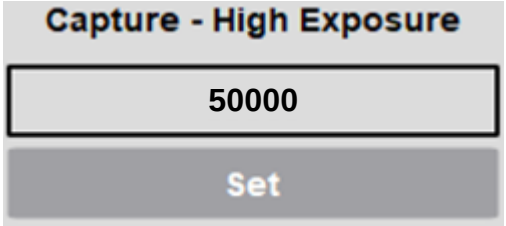
1 Set Display Exposure

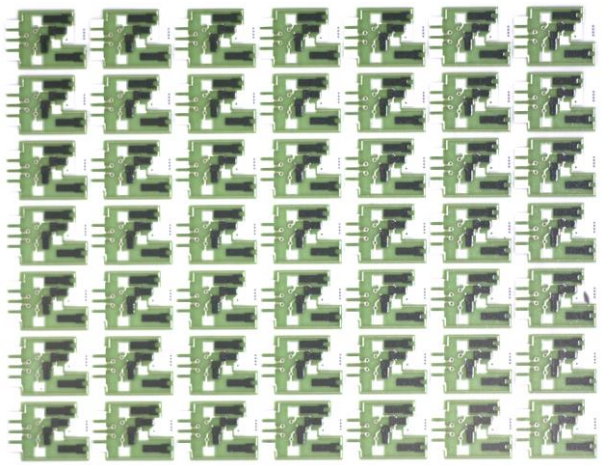
This is the exposure which will bring the normal and correct brightness



2 Set High Exposure

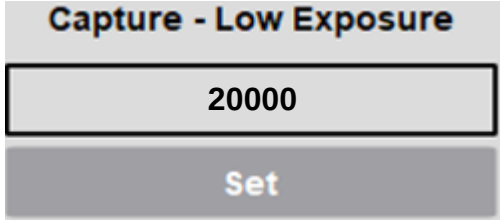
This is to simulate the possible brightness variation which will cause the brighter and clear image

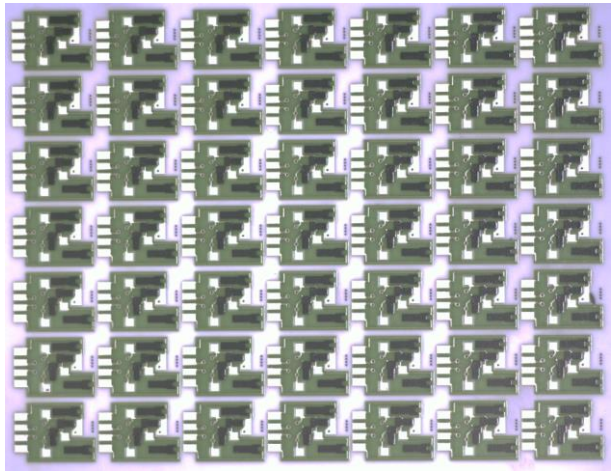




3 Set Low Exposure

This is to simulate the possible brightness variation which will cause the darker and clear image



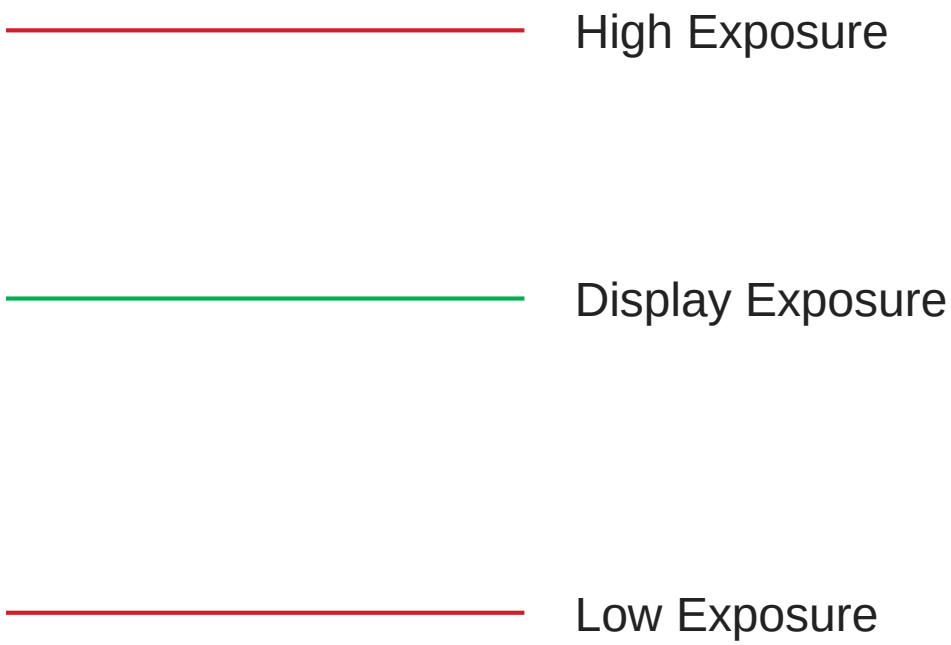


TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Set Correct Value for “Display Exposure”, “High Exposure”, “Low Exposure”?

Case 1:
If the lighting variation is small in production line, for example, the whole lighting environment is covered by the dark glass or mask, the high exposure and low exposure can be set very close to display exposure

Case 2:
If the lighting variation is big in the production line, for example the ambient light will influence the brightness, the high exposure and low exposure can be set with high deviation to display exposure

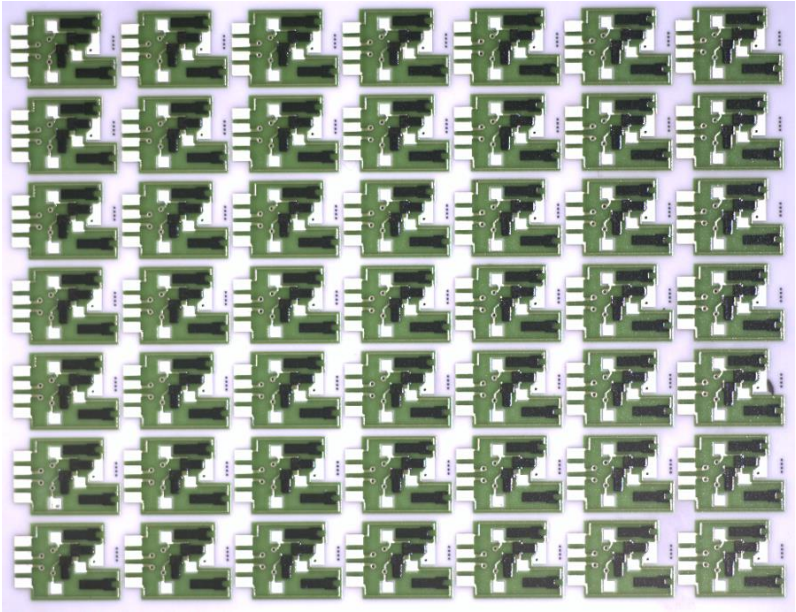


TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

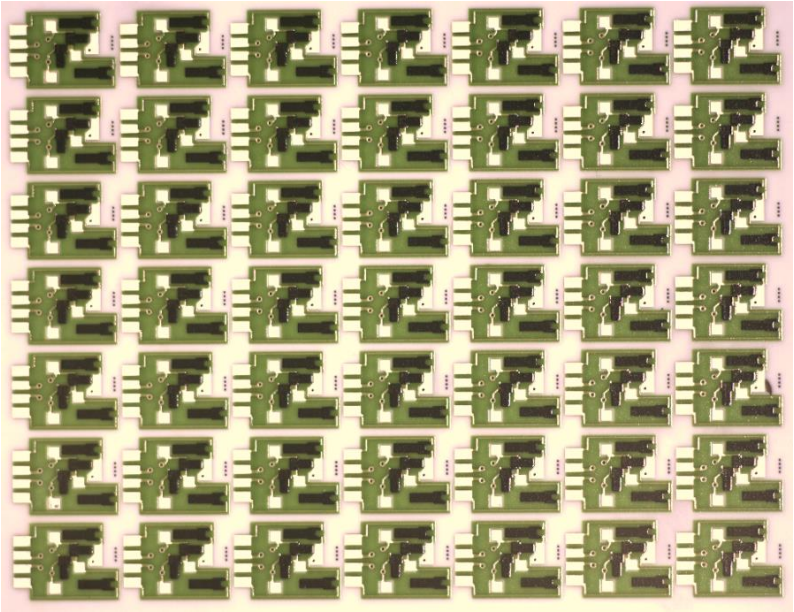
Do I Need to Perform the White Balance?

Perform the white balance in the following conditions

- Use the camera first time to capture the image
- Aperture change
- Exposure change
- Lighting intensity change
- Ambient lighting environment change
- If you product image has the color distortion, white balance is needed



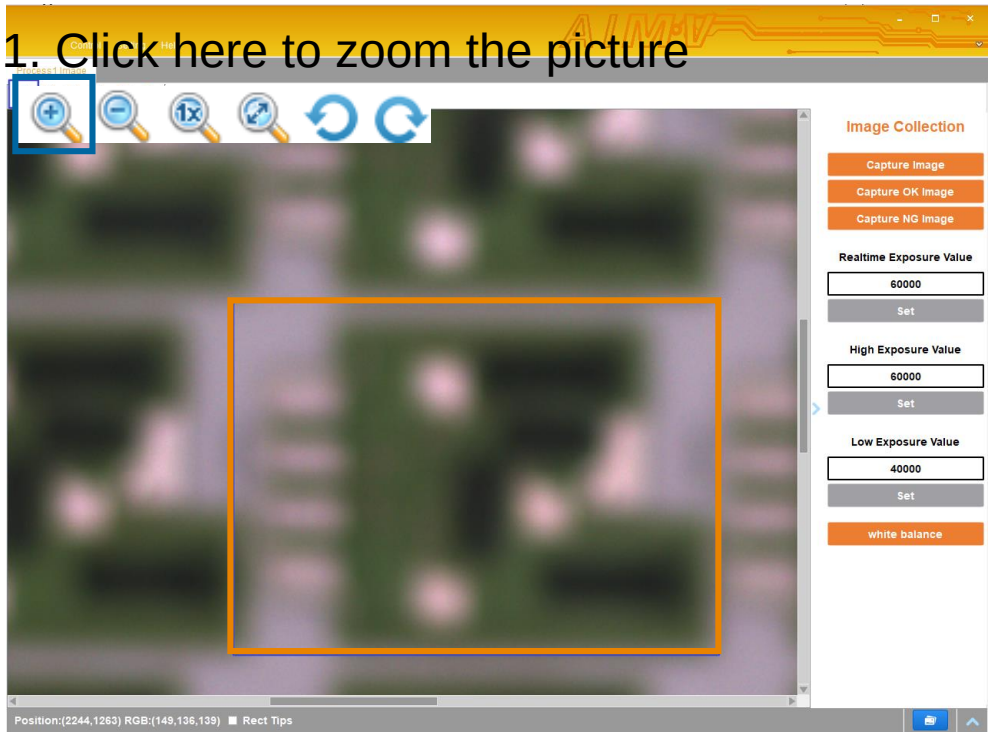
Correct Color of the Product Image



Color Distortion of the Product Image

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Acquire Images with Good Quality for Training?



Rotate here to adjust the height of the camera

+

Rotate here to adjust the lens focus

1. Click the zoom button to have **a zoomed view containing only one product**
2. Adjust the height of the camera and/or the lens to focus on the product

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

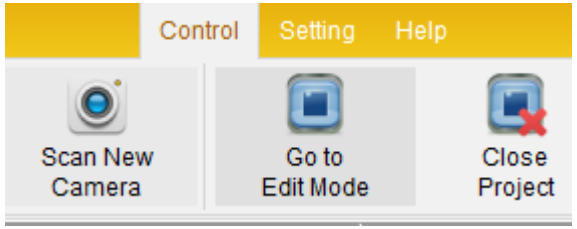
How to Acquire Images with Good Quality for Training?

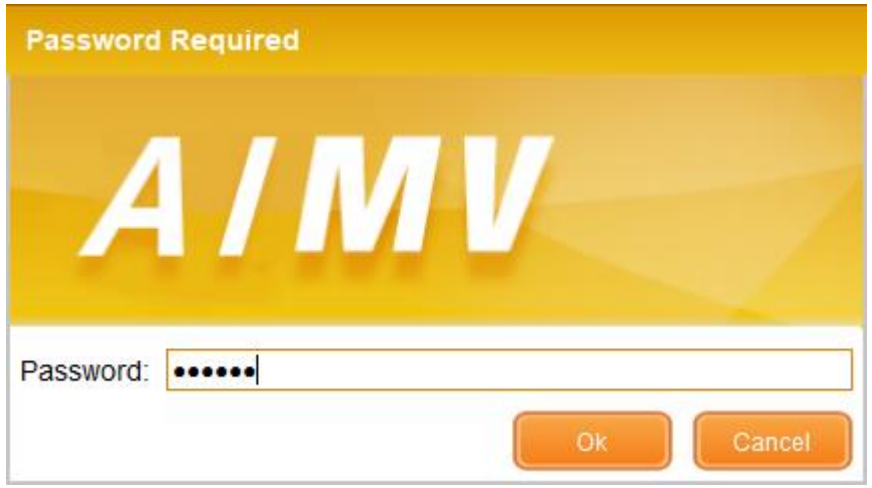
3. You get a clear image

- Make sure the lighting condition is:
  - Even. Lighting effects on product side and center look even
  - No reflection
  - With proper brightness so the main features are clearly visualized
- Make sure the defect in the image can be clearly seen

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Stop Real-time Display During Inline Image Collection?



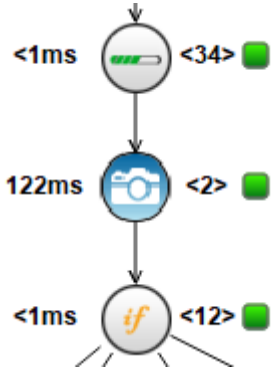


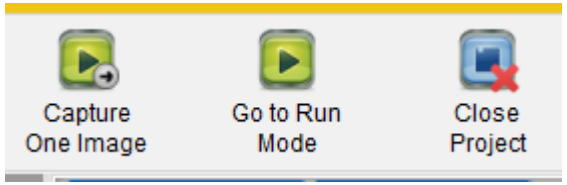
1. Click the “Go to Edit Mode” button

2. Input 123456 and click “Ok” button

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Stop Real-time Display During Inline Image Collection?





3. Click node

4. Switch “Trigger” to “External Trigger”, and input “RealTime” in the “Variable”

5. Click the “Go to Run Mode” button to go to Run Mode

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

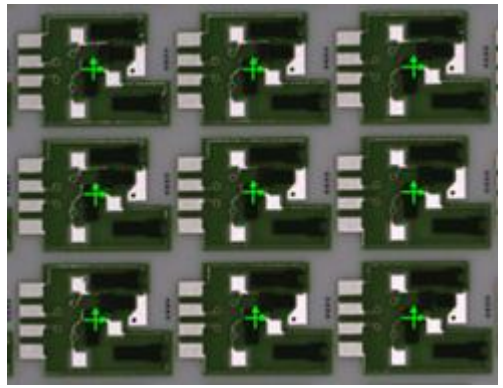
Return to Contents

AI Defect Inspection – Step 2 for Image Segmentation

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

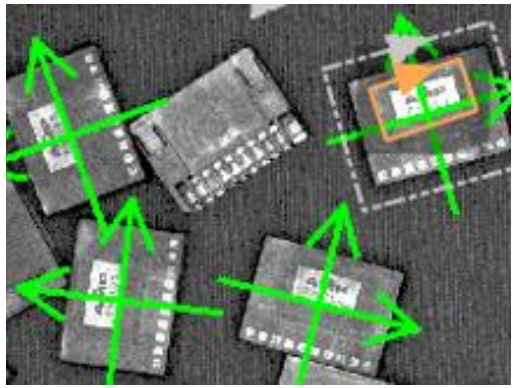
Does My Application Need Image Segmentation?

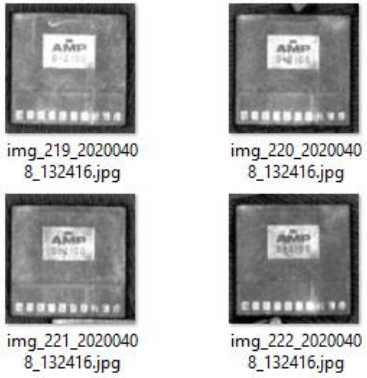
Image Segmentation is needed for following three situations



Situation 1
Many Product within One Image

Situation 2
Remove the Background



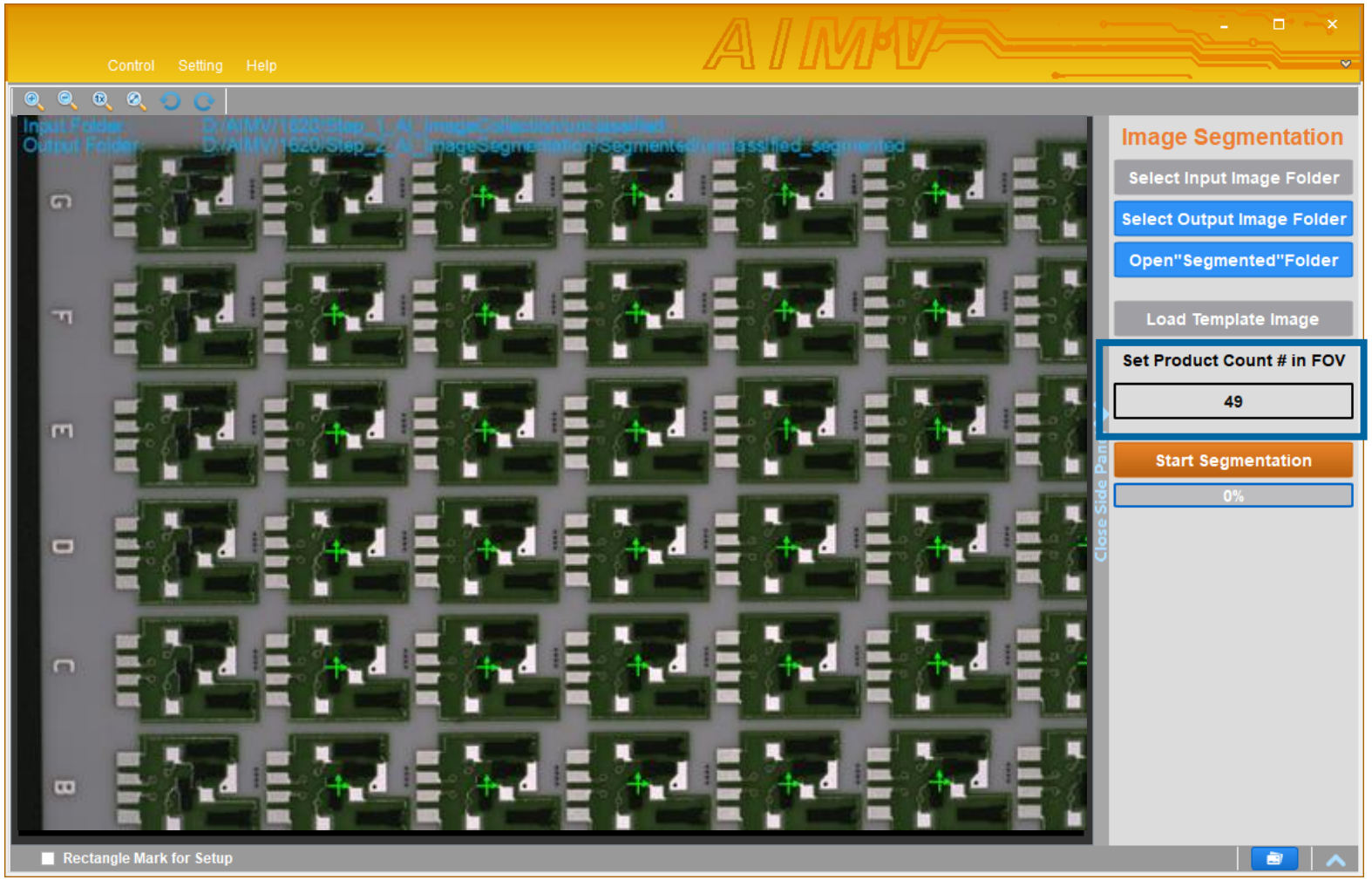


Situation 3
Rotated Product

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Why can’t Segment All the Products Within One Image? (For Aligned Product)

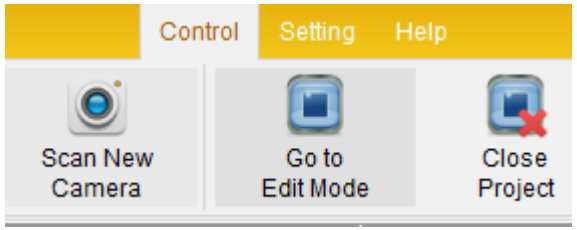
- Make Sure the Product Count # in FOV is set correctly



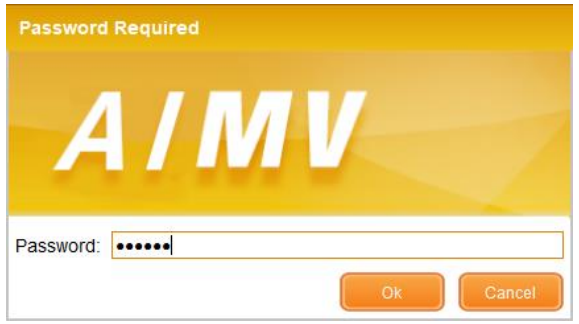
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Why can't Segment All the Products Within One Image? (For Aligned Product)

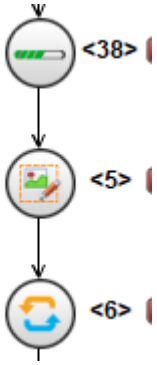
- Check the Segmentation Threshold and Segmentation Method



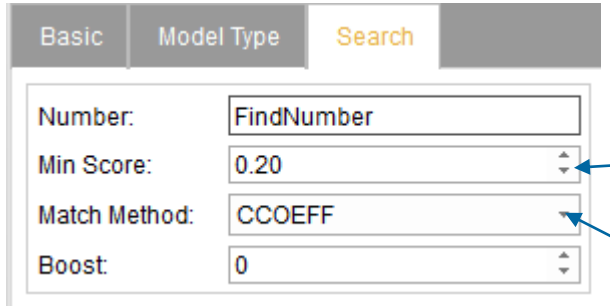
1. Click the "Go to Edit Mode" button



2. Input 123456 and click "Ok" button

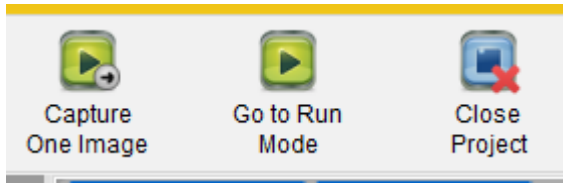


3. Click node



- Min Score: the smaller the value is, the easier to find the product
- Match Method: different segmentation method

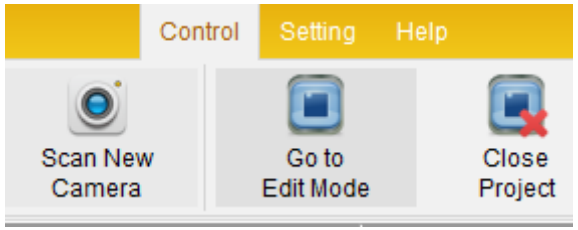
4. Switch to "Search" tab, and change "Min Score" and "Match Method"



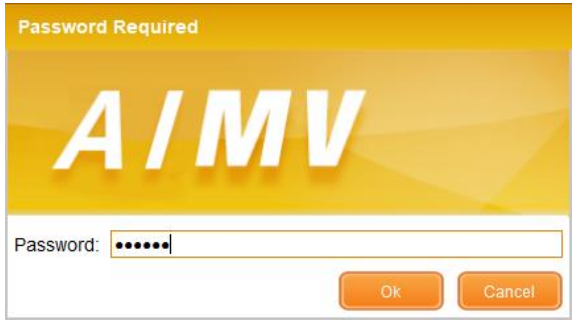
5. Click the "Go to Run Mode" button to go to Run Mode and test the effect

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

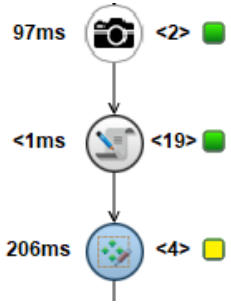
Why can't Segment All the Products Within One Image? (For Rotated Product)



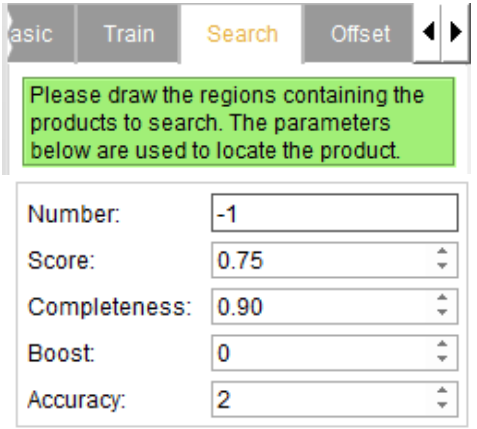
1. Click the "Go to Edit Mode" button



2. Input 123456 and click "Ok" button

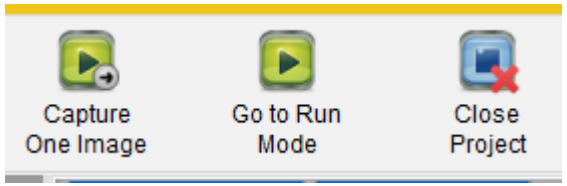


3. Click node



3. Switch to "Search" tab. Change "Number", "Score" and "Completeness"

- Number: the product number to find, -1 means find all possible products
- Score: control the similarity between found product and trained template. 0.75 means at least 75% similarity is required for one found product
- Set "Completeness" to control the completeness of one found product. 0.90 means at least 90% completeness



6. Click the "Go to Run Mode" button to go to Run Mode and test the effect

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

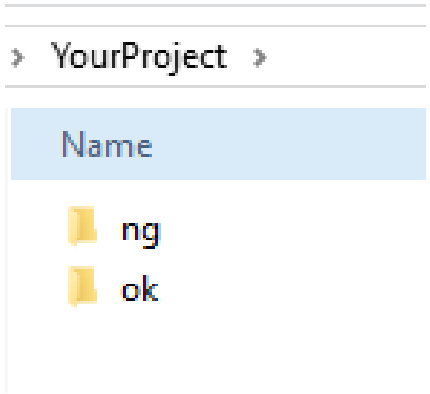
AI Defect Inspection – Step 3 for Image Labelling

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Rule for Image Labelling?

Before labelling:

- Make sure image segmentation is done
- Make sure each picture only contains one product and the product is complete



Labelling:

- Create two folders named "ok" and "ng"
- Navigate into the folder for the segmented pictures
- Check each segmented picture and put it in "ok" or "ng" folder
- If one sample can not easily be decided as ok, it shall be put into ng without hesitance

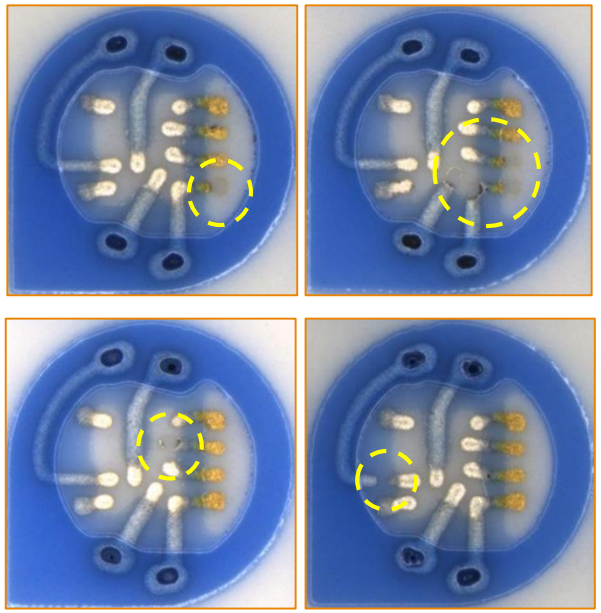
After labelling:

- Make sure each picture in "ok" folder is for "ok" product, and each picture in "ng" folder is for "ng" product
- Make sure the image quantity is not less than the recommended number in Table for Image Sample Size Evaluation Slide
- Make sure the ratio of "ng" samples > 20% of total sample. For example, if there are 200 pictures in "ok" folder, 100 pictures in "ng" folder, then the ratio of "ng" samples is 100/(200+100) = 33.3%, which is > 20%
- If the ratio of "ng" samples is < 20%, collect more "ng" samples

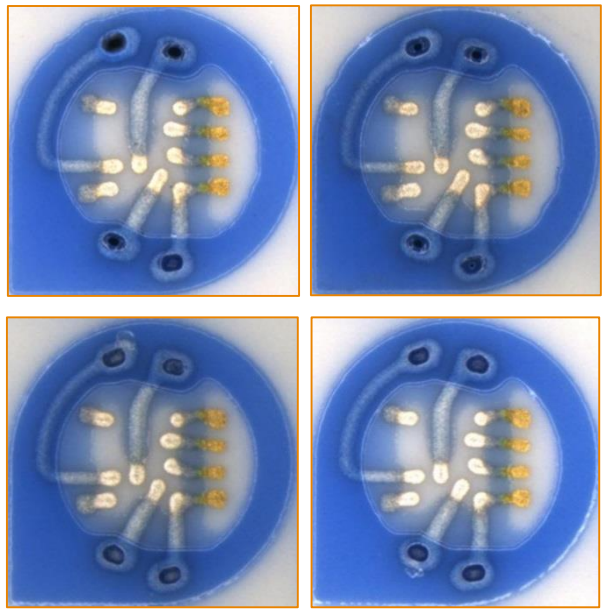
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What’s the Rule for Image Labelling?

- To make sure the AI model accuracy, the labelled “ok” folder shall include all typical “ok” products, and the labelled “ng” folder shall include all “ng” products
- The criteria between ok and ng shall be clear and definite
- If one sample can not easily be decided as ok, it shall be put into ng without hesitance



NG Category



OK Category

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AI Defect Inspection – Step 3 for Image Marking

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What’s the Rule for Image Marking?

If you want to use Defect Location Runtime Tool, you have to mark images with this tool. If you just use Classification Runtime Tool, then you can skip this part

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Return to Contents

AI Defect Inspection – Step 4 for Train Model by AWS Server

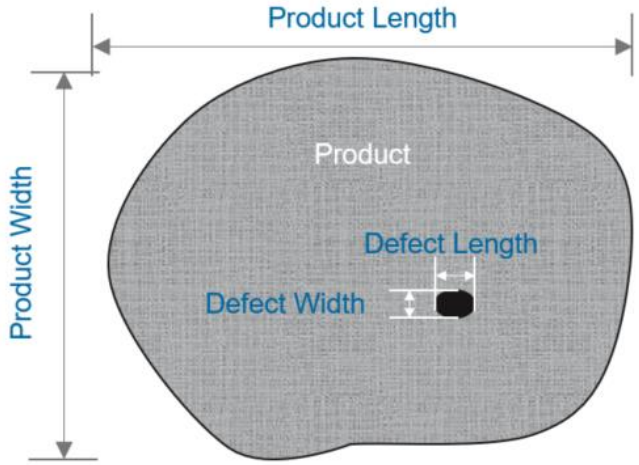
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What's the Relationship Between Product Volume and the Accuracy for Classification?

- The suggested image quantity to train the AI model can be found through the following table:

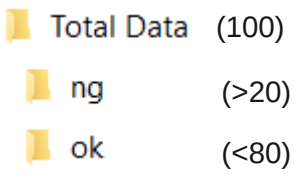
Table for Image Sample Size Evaluation

Expected Accuracy \ Defect Area Rate	2%	3%	4%	5%	10%	20%	30%	80%
99.9%	4706	1857	581	289	72	36	28	21
99.5%	4602	1823	573	286	71	36	28	21
99.0%	4475	1781	563	282	71	36	28	21
98.0%	4232	1700	544	274	70	35	28	21
97.0%	4002	1623	525	267	69	35	28	21



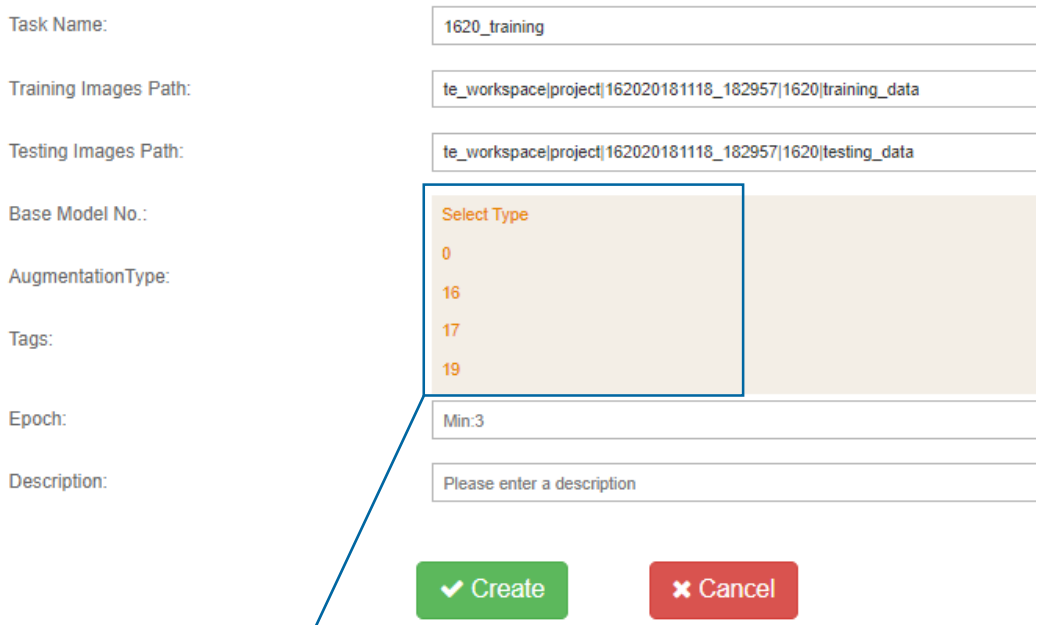
Defect Area Rate =
Defect Length * Defect Width/(Product Length*Product Witch)

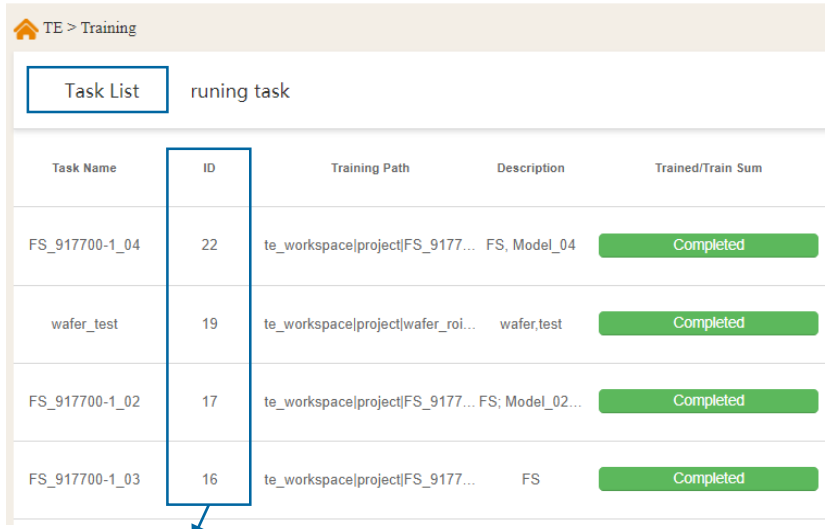
- For example, if expected inspection accuracy is 99.9%, and defect area rate to the whole product is 3% (please see the up-right figure). Please look up the column of 3% and the row of 99.9%, the suggested image quantity is 1857
- If your input values exceed the table range, please refer to the maximum size 4706
- Please make sure NG image quantity of the total image quantity is > 20%. For example, if total image quantity is 100, so the ng image quantity should be larger than 20, as shown in the right figure



TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Train One Model Based on One Existing Model?





Here are the “Base model No.” trained in previous training tasks

Choose Base Model No. in the list. For the first time training, select “0”. For the **incremental training**, select the base model No. you want to reuse

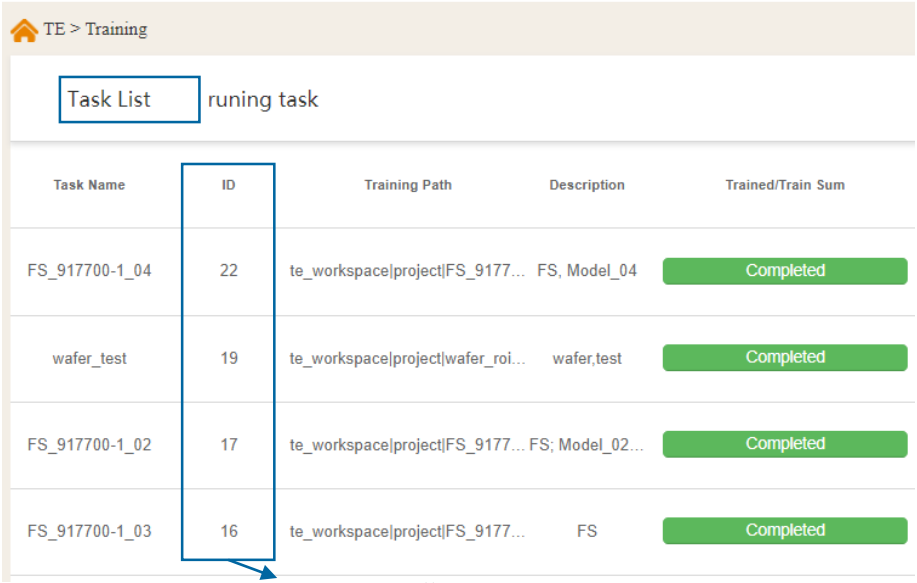
Incremental training: when you have trained an AI model but you are not satisfied with the model accuracy, and you want to further train the AI model to get a better accuracy, you can create a new training task by selecting the Base Model No. based on which you want to further train the AI model. This will save the training time instead of training from scratch

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Train One Model Based on One Existing Model?

For **incremental training:**

- Add newly collected images into the original *training_data* folder and repackage the folder
- Select the base model No. you want to use, instead of input “0”

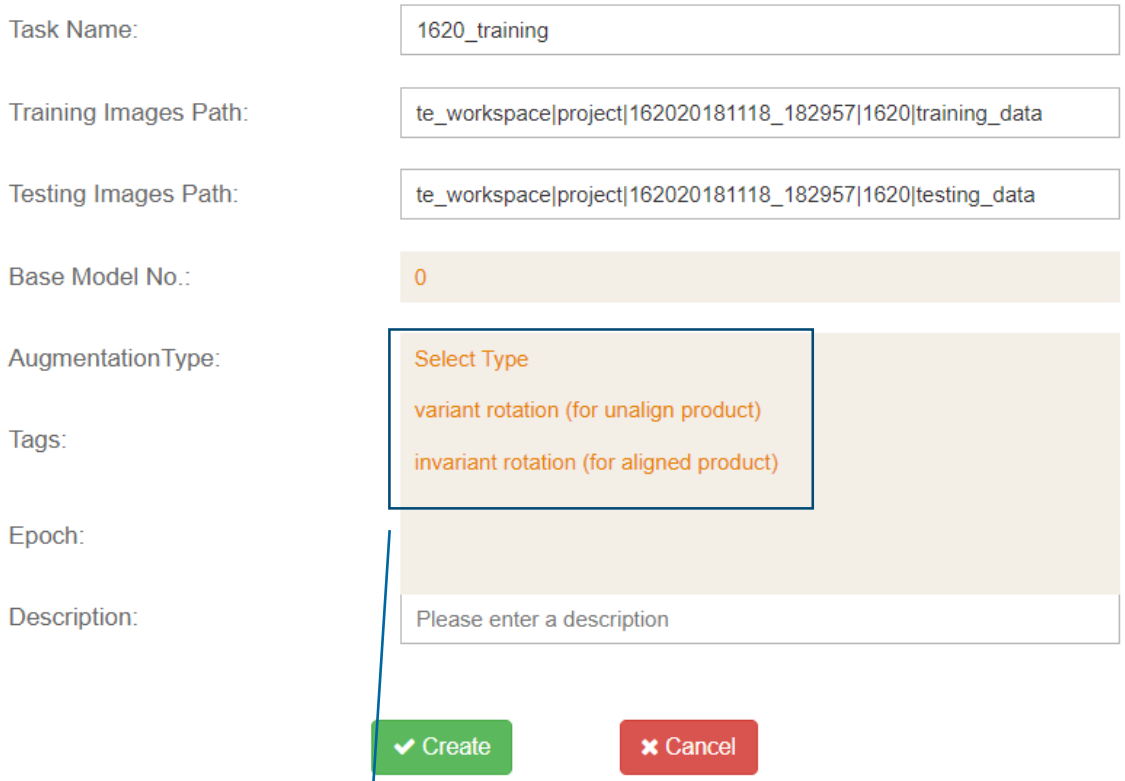


Here are the “Base model No” trained in previous training tasks

For example, if you have trained an AI model, the task ID of which is 22, you can collect more new “ok” and “ng” images and add these new images into the original *training_data* folder. Then zip the new *training_data* folder with the original *testing_data* folder together and upload the zip file. When creating a new task, you shall select the Base Model No. 22. This is an incremental training process.

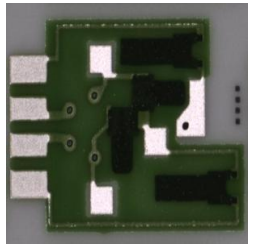
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

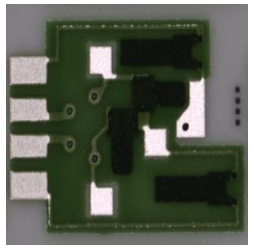
How to Train One Model Applied for Unaligned Product? (Classification Only)



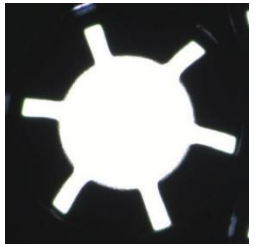
select “variant rotation (for unalign product)”

Align product: The product’s orientation in the image is the same
Unalign product: The product’s orientation in the picture may be rotated





Aligned product



Unalign product

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What to Do If the AI Classification Accuracy is Not Good Enough?

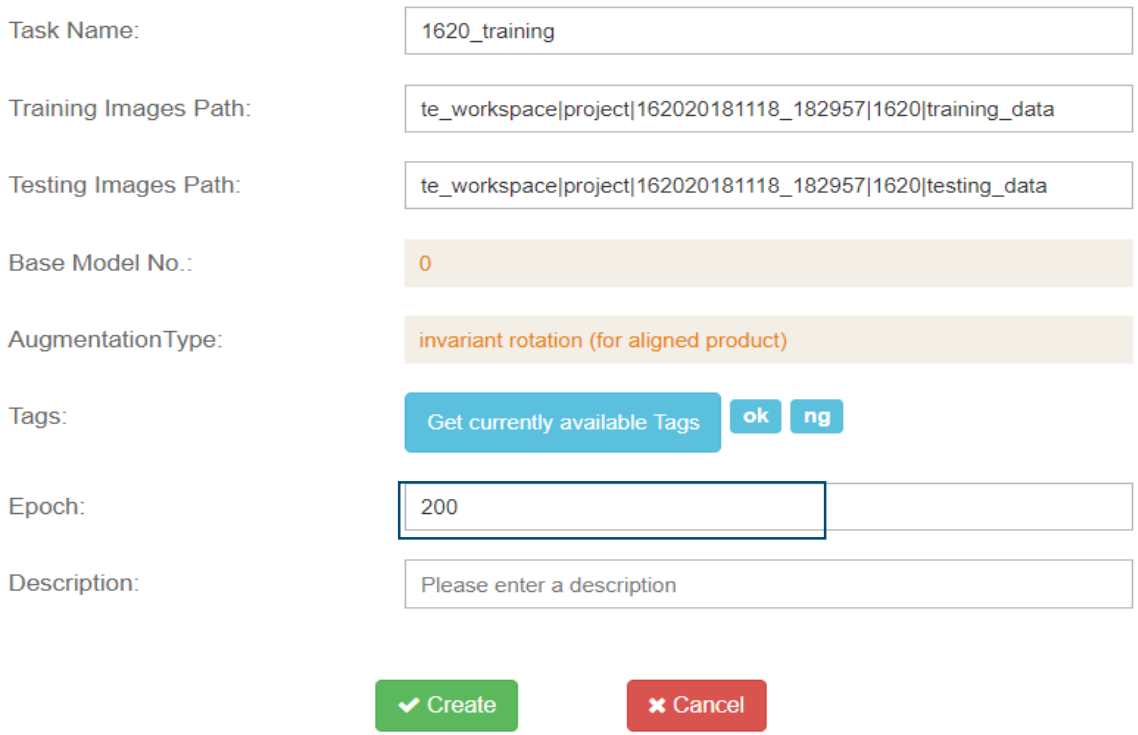
Tips to improve the accuracy:

- Clean the training samples
  - Review pictures in ok and ng folder, make sure the pictures in ok folder are all *ok* products, the pictures in ng folder are all *ng* products
  - Remove pictures which are not easy to decide whether they are ok or ng from the training folder

- Collect more new samples to add into the training set. Make sure the ratio of ng samples > 20% of total sample

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Decide Epoch Number?



Epoch indicates the training time to be consumed

- For first time training, input 3 for one quick run test. If the training is completed successfully, increase the epoch number
- Input 100 as default for quick assessment
- It is recommended to input 200 for precise AI model training

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Extend the AI Model Training Hours for AWS Training Platform?

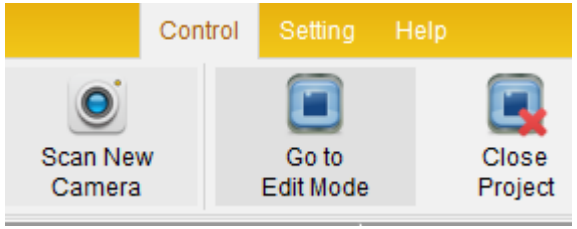
Please contact james.yu@te.com

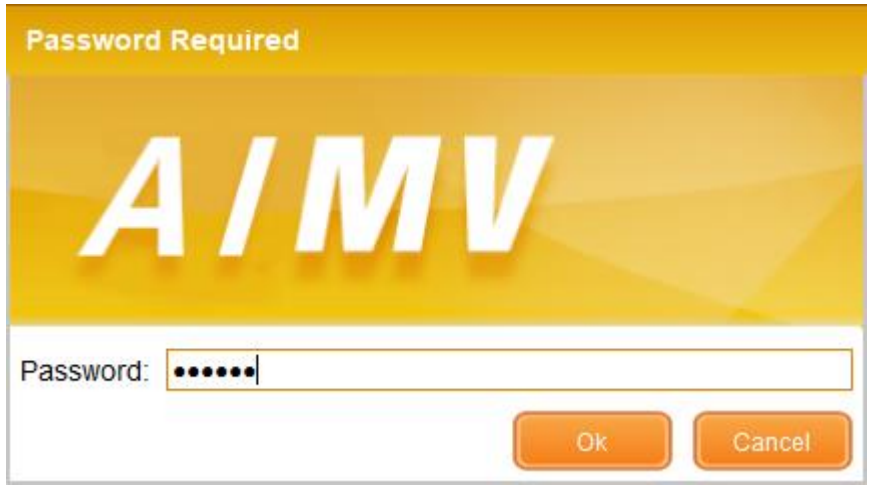
TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

AI Defect Inspection – Step 5 for Deployment

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Stop Real-time Display During Inline Deployment?



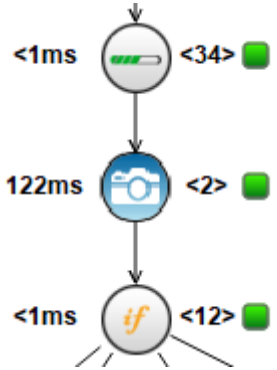


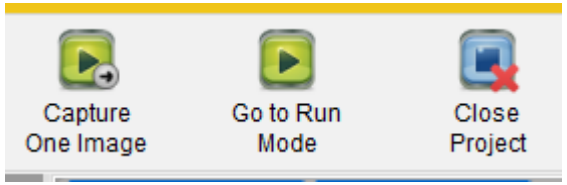
1. Click the “Go to Edit Mode” button

2. Input 123456 and click “Ok” button

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

How to Stop Real-time Display During Inline Deployment?





3. Click node

4. Switch “Trigger” to “External Trigger”, and input “RealTime” in the “Variable”

5. Click the “Go to Run Mode” button to go to Run Mode

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

What to Do If Need a Customized GUI?

- please contact us for technical support

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

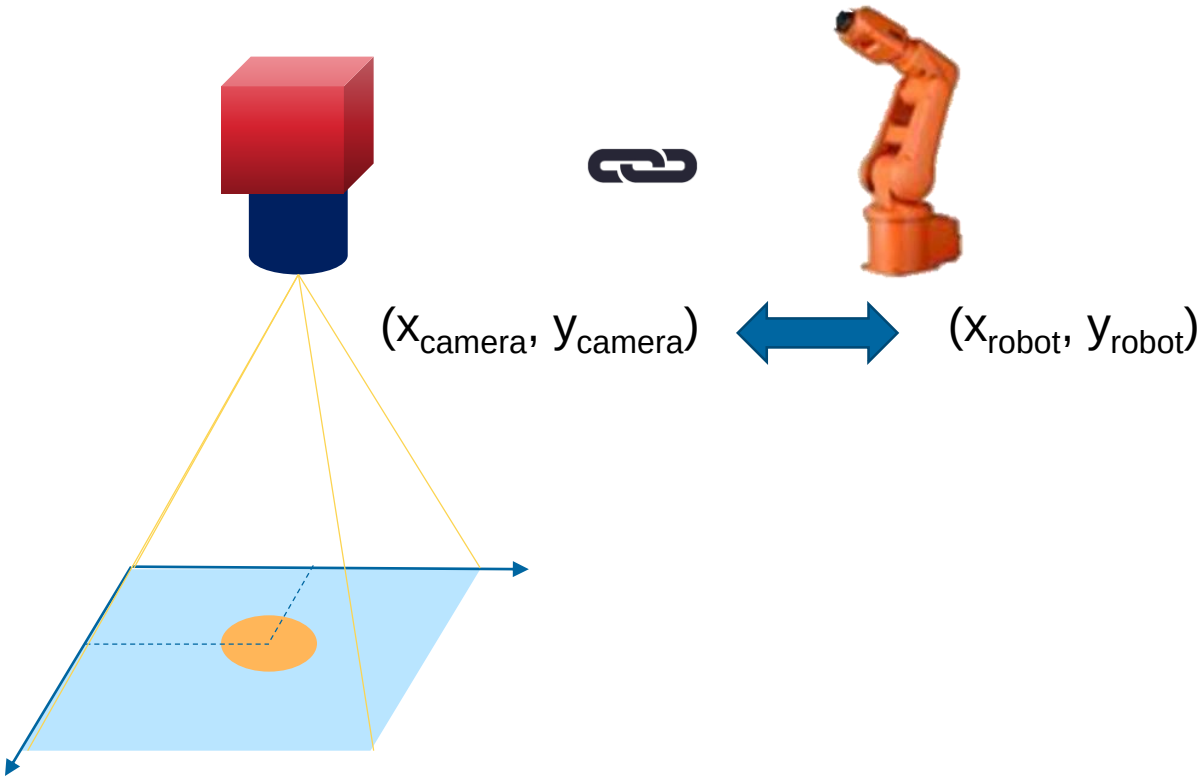
Calibration

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Calibration Introduction: What's Calibration?

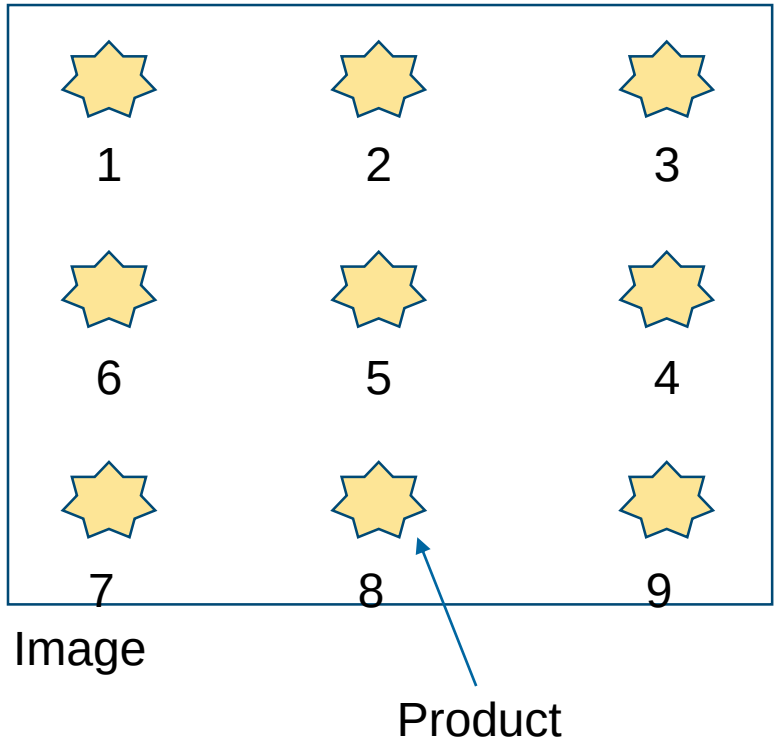
- Calibration means to define the conversion between robot coordinate and camera coordinate.

- For example, the camera locates the position of one product, the result is the camera coordinate (x_{camera}, y_{camera}). The robot needs the robot coordinate (x_{robot}, y_{robot}) of the product. After calibration, the AIMV software will convert camera coordinate (x_{camera}, y_{camera}) to robotic coordinate (x_{robot}, y_{robot})



 TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Calibration Introduction: How to Define the Conversion Relationship?



1. Place 9 products within one image, the 9 products should be placed in orthogonal 3 cols by rows
2. Get the camera coordinates ad robot coordinates of the nine products
3. Calculate the conversion relationship between camera and robot
4. The Conversion Relationship is one matrix

Product	Camera Coordinate	Robot Coordinate
1	(x_{c1}, y_{c1})	(x_{r1}, y_{r1})
2	(x_{c2}, y_{c2})	(x_{r2}, y_{r2})
3	(x_{c3}, y_{c3})	(x_{r3}, y_{r3})
4	(x_{c4}, y_{c4})	(x_{r4}, y_{r4})
5	(x_{c5}, y_{c5})	(x_{r5}, y_{r5})
6	(x_{c6}, y_{c6})	(x_{r6}, y_{r6})
7	(x_{c7}, y_{c7})	(x_{r7}, y_{r7})
8	(x_{c8}, y_{c8})	(x_{r8}, y_{r8})
9	(x_{c9}, y_{c9})	(x_{r9}, y_{r9})

Calibrate Conversion Relationship

$$\begin{bmatrix} m_1 & m_2 & m_3 \\ m_4 & m_5 & m_6 \\ m_7 & m_8 & m_9 \end{bmatrix}$$

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

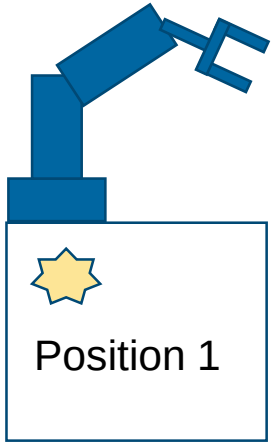
Calibration Introduction: Important Tip

- **After Calibration, please do not move the camera, lens and lighting unit.**
- **If any of them is moved, please perform calibration again**

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Autocalibration: Introduction

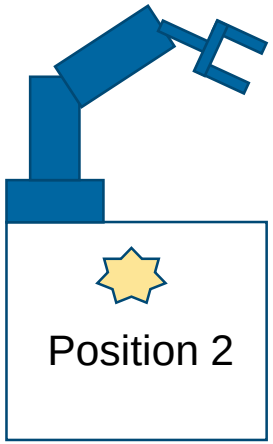
- Automatic Calibration means the robot will send the coordinate of the product to AIMV software, no need to use the teach pendant of Robot to manually get the product's robot coordinate.



Step 1:

1. Robot places the product in position 1
2. Robot triggers the AIMV software to capture one image, locate the position of the product. and get the camera coordinate
3. Robot sends the robot coordinate of position 1

Product	Camera Coordinate	Robot Coordinate
1	(x_{c1}, y_{c1})	(x_{r1}, y_{r1})



Step 2:

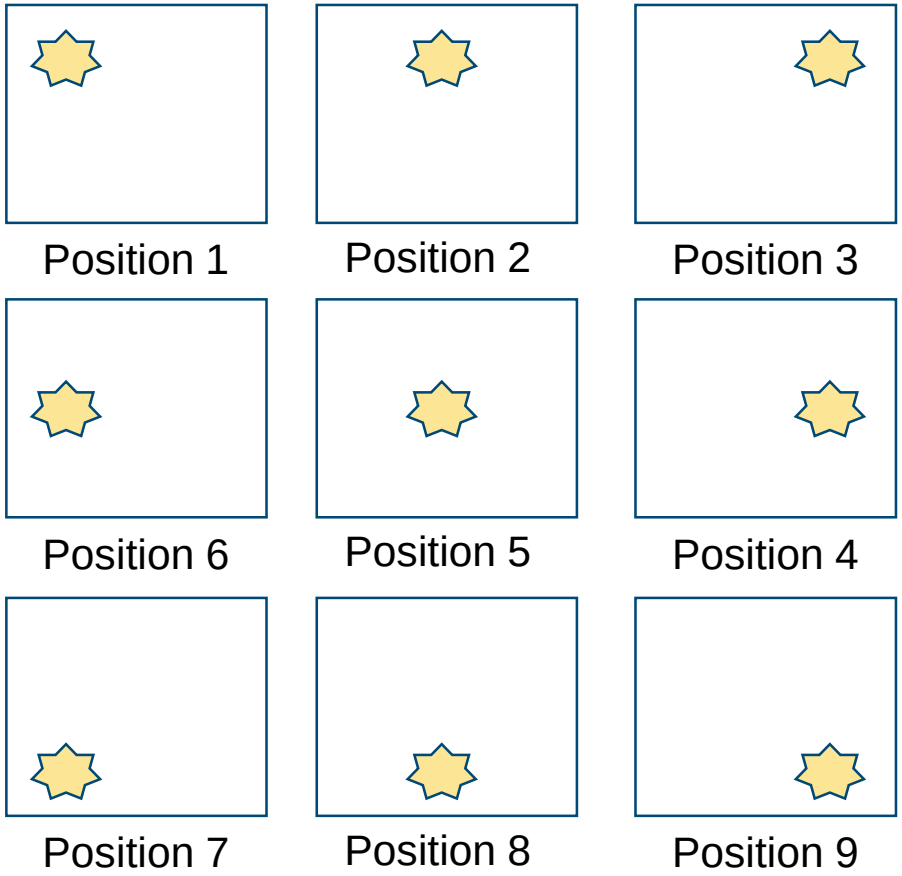
1. Robot places the product in position 2
2. Robot triggers the AIMV software to capture one image, locate the position of the product. and get the camera coordinate
3. Robot sends the robot coordinate of position 2

Product	Camera Coordinate	Robot Coordinate
1	(x_{c1}, y_{c1})	(x_{r1}, y_{r1})
2	(x_{c2}, y_{c2})	(x_{r2}, y_{r2})

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Autocalibration: Introduction

Repeat the previous step 9 times, we will get the calibration done



Product	Camera Coordinate	Robot Coordinate
1	(x_{c1}, y_{c1})	(x_{r1}, y_{r1})
2	(x_{c2}, y_{c2})	(x_{r2}, y_{r2})
3	(x_{c3}, y_{c3})	(x_{r3}, y_{r3})
4	(x_{c4}, y_{c4})	(x_{r4}, y_{r4})
5	(x_{c5}, y_{c5})	(x_{r5}, y_{r5})
6	(x_{c6}, y_{c6})	(x_{r6}, y_{r6})
7	(x_{c7}, y_{c7})	(x_{r7}, y_{r7})
8	(x_{c8}, y_{c8})	(x_{r8}, y_{r8})
9	(x_{c9}, y_{c9})	(x_{r9}, y_{r9})

Conversion Relationship

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Terminology

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Terminology List

Click to See Definition

- Exposure
- Working Distance
- Telecentric Lens
- FA Lens
- Focal Length

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Contact Us

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Return to Contents

Contact Us

AMT Contact

AMT Shanghai:

Alex Zhou, zhou_alex@te.com

Tel: +86-021-33259280

AMT Harrisburg:

Sonny O Osunkwo, sonny.osunkwo@te.com

Tel: +1-7179865673

AMT Hermosillo:

Angel A Slistan, angel.slistan@te.com

Tel: +52-6621432067

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.

Thanks

TE Connectivity Confidential & Proprietary. Do not reproduce or distribute.